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(54) CONTROLLED INFORMATION FLOW FROM TEACHER MODEL TO STUDENT MODEL FOR KNOWLEDGE DISTILLATION AND TRANSFER

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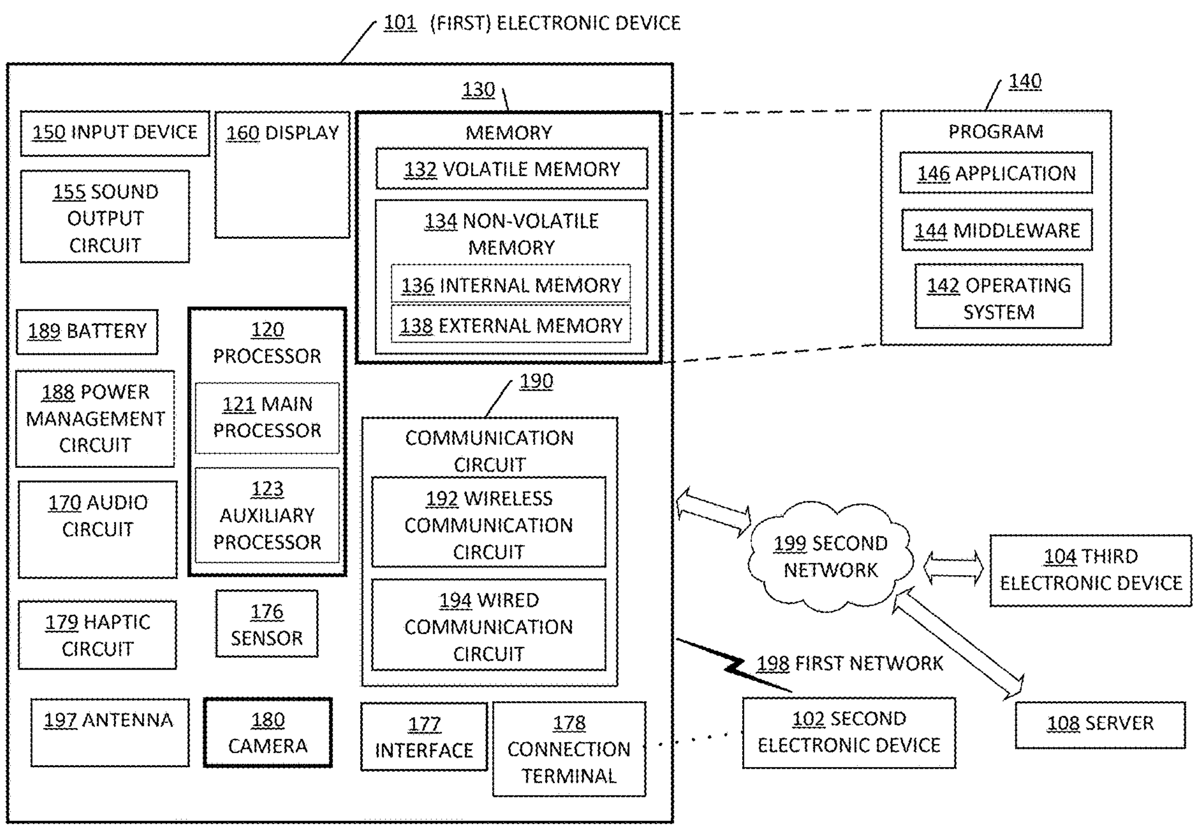
(57) **ABSTRACT**

A computer-implemented method performed by an electronic device configured to distill knowledge from a teacher model and transfer the distilled knowledge to a student model, includes: obtaining training data; obtaining the teacher model; training at least one rate distortion module (RDM) that is not a teacher assistant (TA) model; training the student model using the trained at least one RDM; and transmitting the trained student model to a first electronic device.



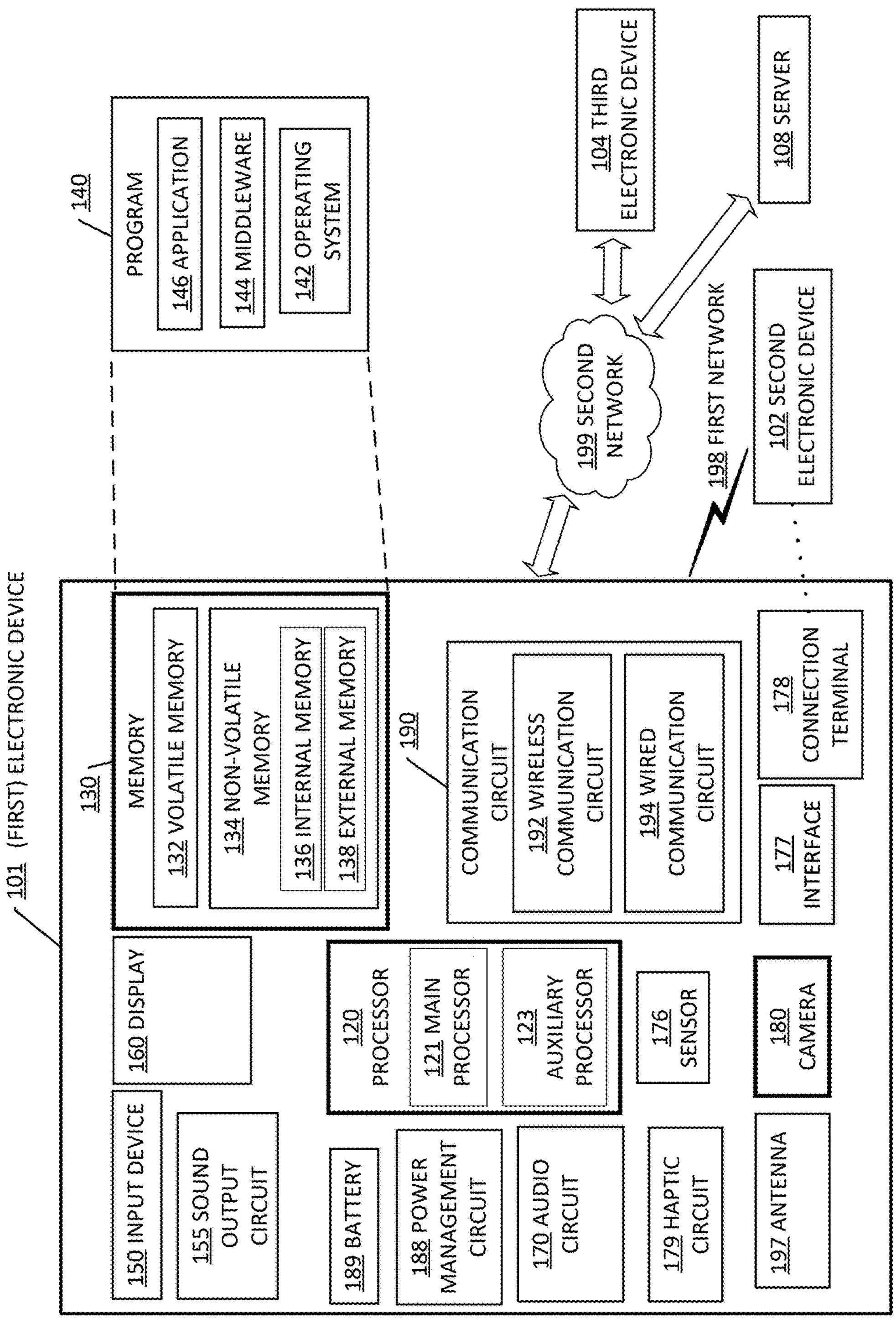

101 (FIRST) ELECTRONIC DEVICE
130
MEMORY
132 VOLATILE MEMORY
134 NON-VOLATILE MEMORY
136 INTERNAL MEMORY
138 EXTERNAL MEMORY
140
PROGRAM
146 APPLICATION
144 MIDDLEWARE
142 OPERATING SYSTEM
104 THIRD ELECTRONIC DEVICE
108 SERVER
199 SECOND NETWORK
198 FIRST NETWORK
102 SECOND ELECTRONIC DEVICE
190
COMMUNICATION CIRCUIT
192 WIRELESS COMMUNICATION CIRCUIT
194 WIRED COMMUNICATION CIRCUIT
178 CONNECTION TERMINAL
177 INTERFACE
160 DISPLAY
120 PROCESSOR
121 MAIN PROCESSOR
123 AUXILIARY PROCESSOR
176 SENSOR
180 CAMERA
150 INPUT DEVICE
155 SOUND OUTPUT CIRCUIT
189 BATTERY
188 POWER MANAGEMENT CIRCUIT
170 AUDIO CIRCUIT
179 HAPTIC CIRCUIT
197 ANTENNA

FIG. 1

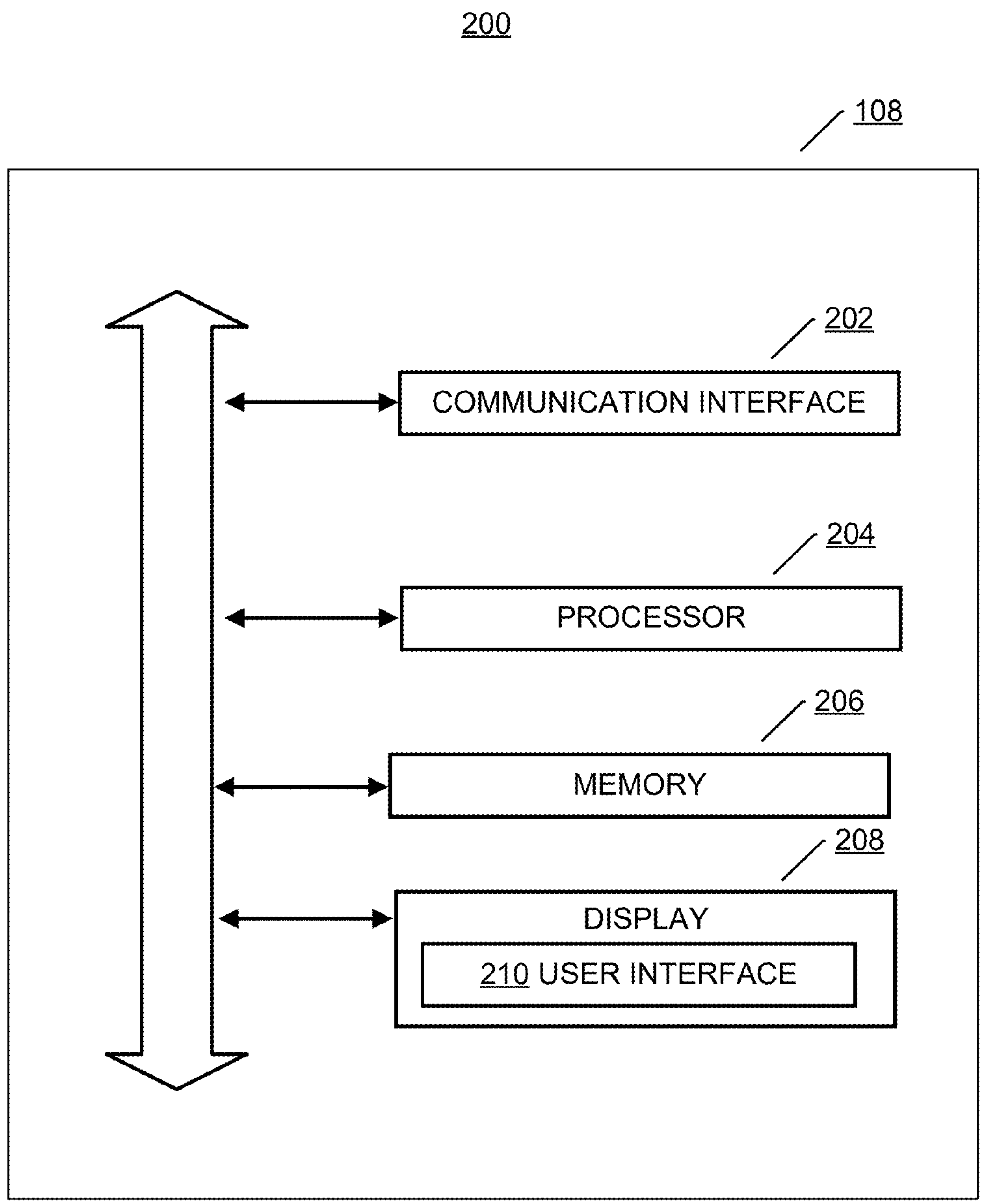

200
108
202
COMMUNICATION INTERFACE
204
PROCESSOR
206
MEMORY
208
DISPLAY
210 USER INTERFACE

FIG. 2

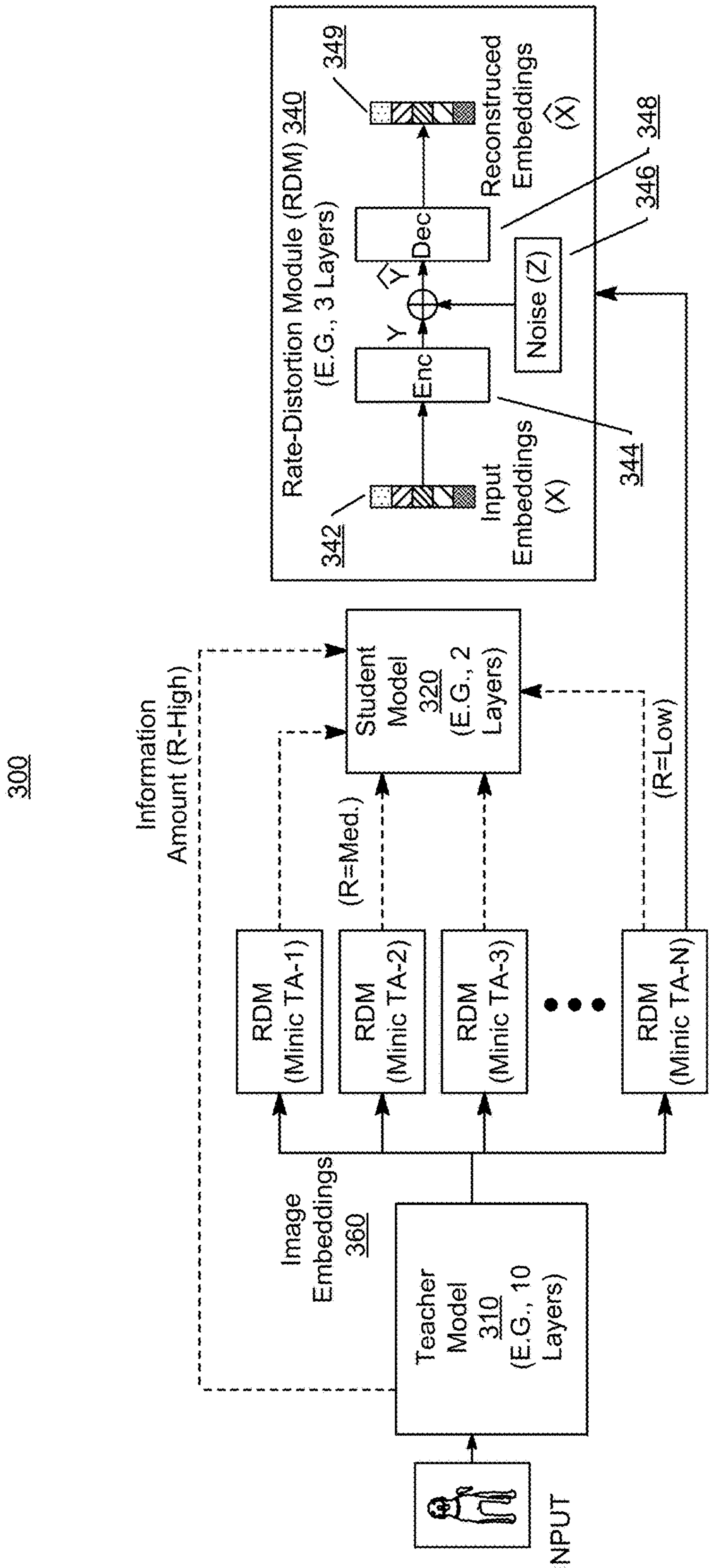

300
INPUT
Teacher Model 310 (E.G., 10 Layers)
Image Embeddings 360
Information Amount (R-High)
RDM (Minic TA-1)
RDM (Minic TA-2)
RDM (Minic TA-3)
RDM (Minic TA-N)
340
(R=Med.)
(R=Low)
Student Model 320 (E.G., 2 Layers)
Rate-Distortion Module (RDM) 340 (E.G., 3 Layers)
342
Input Embeddings (X)
Enc
344
Y
Noise (Z)
346
Ŷ
Dec
348
349
Reconstruced Embeddings (X̂)

FIG. 3

400

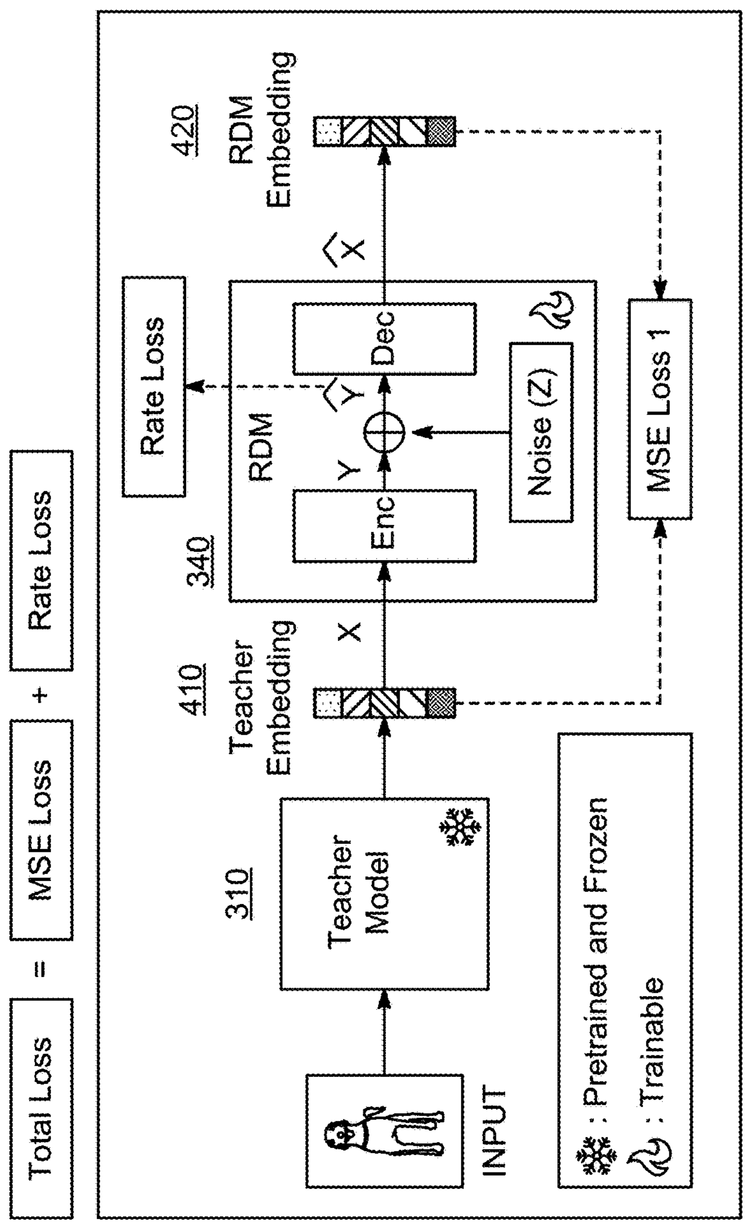

Total Loss = MSE Loss + Rate Loss
INPUT
310
Teacher Model
410
Teacher Embedding
X
340
RDM
Enc
Y
Noise (Z)
Dec
Ŷ
Rate Loss
X̂
420
RDM Embedding
MSE Loss 1
: Pretrained and Frozen
: Trainable

FIG. 4

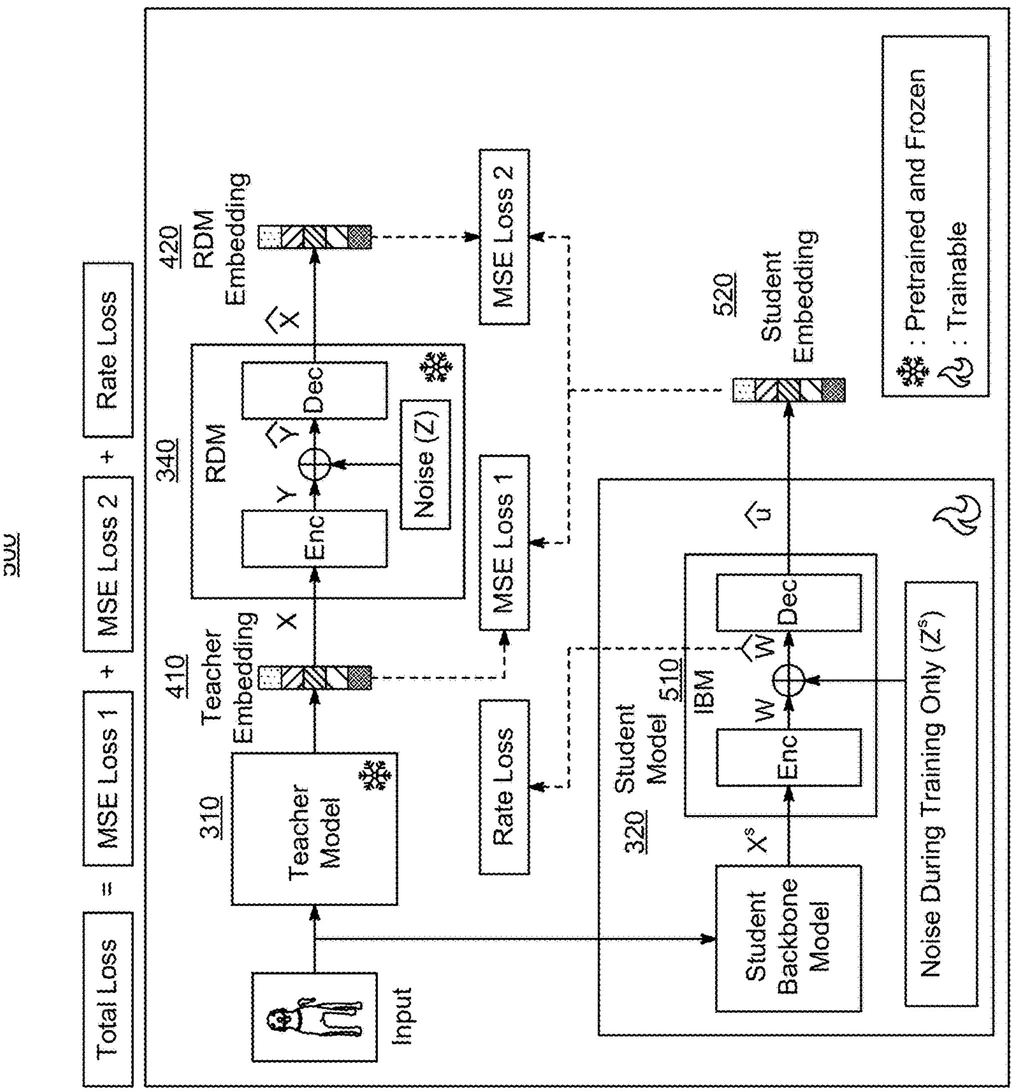
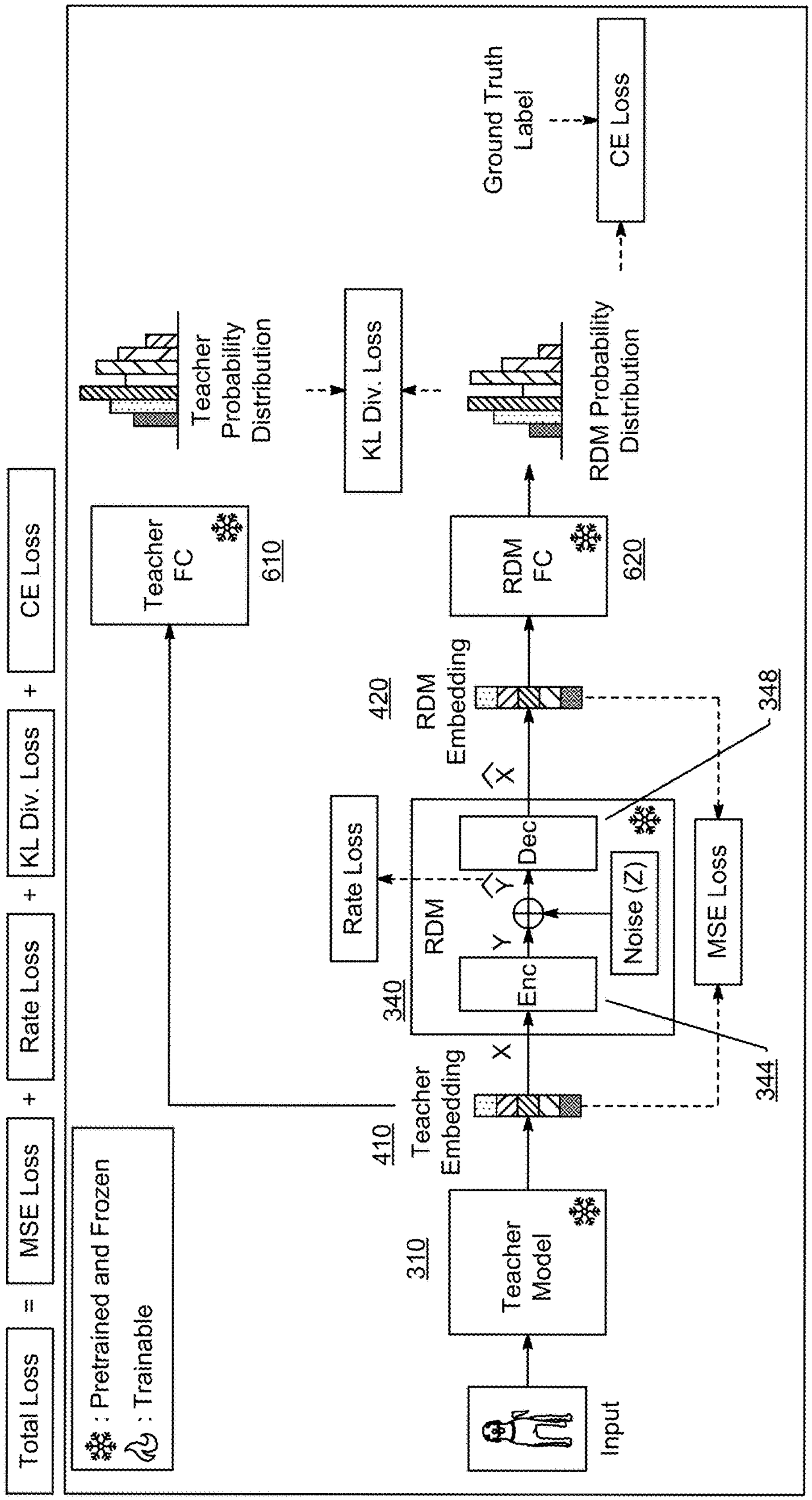

500
Total Loss
=
MSE Loss 1
+
MSE Loss 2
+
Rate Loss
Input
310
Teacher Model
410
Teacher Embedding
X
340
RDM
Enc
Y
Ŷ
Dec
Noise (Z)
X̂
420
RDM Embedding
MSE Loss 2
MSE Loss 1
Rate Loss
320
Student Model
Student Backbone Model
X^s
510
IBM
Enc
W
Ŵ
Dec
Noise During Training Only (Z^s)
Û
520
Student Embedding
: Pretrained and Frozen
: Trainable

FIG. 5

600



Total Loss
=
MSE Loss
+
Rate Loss
+
KL Div. Loss
+
CE Loss
: Pretrained and Frozen
: Trainable
Input
310
Teacher Model
410
Teacher Embedding
X
340
Enc
Y
RDM
Noise (Z)
Ŷ
Dec
X̂
344
348
Rate Loss
MSE Loss
420
RDM Embedding
Teacher FC
610
RDM FC
620
Teacher Probability Distribution
KL Div. Loss
RDM Probability Distribution
Ground Truth Label
CE Loss

FIG. 6

700

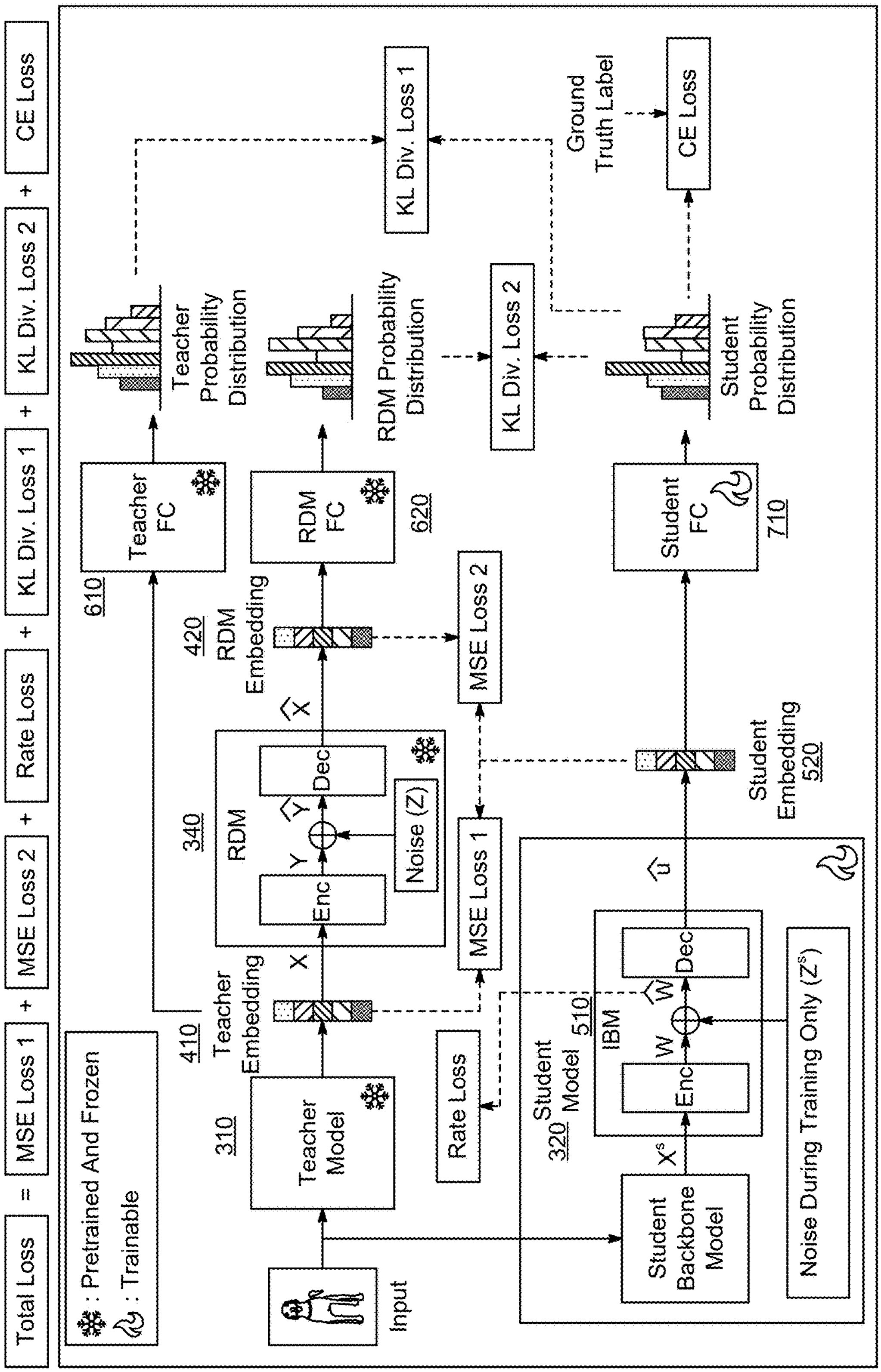

Total Loss = MSE Loss 1 + MSE Loss 2 + Rate Loss + KL Div. Loss 1 + KL Div. Loss 2 + CE Loss
: Pretrained And Frozen
: Trainable
Input
310
Teacher Model
410
Teacher Embedding
X
340
RDM
Enc
Y
Ŷ
Noise (Z)
Dec
X̂
420
RDM Embedding
610
Teacher FC
620
RDM FC
Teacher Probability Distribution
RDM Probability Distribution
KL Div. Loss 1
KL Div. Loss 2
MSE Loss 1
MSE Loss 2
Rate Loss
320
Student Model
Student Backbone Model
X^s
510
IBM
W
Ŵ
Noise During Training Only (Z^s)
û
Student Embedding
520
710
Student FC
Student Probability Distribution
Ground Truth Label
CE Loss

FIG. 7

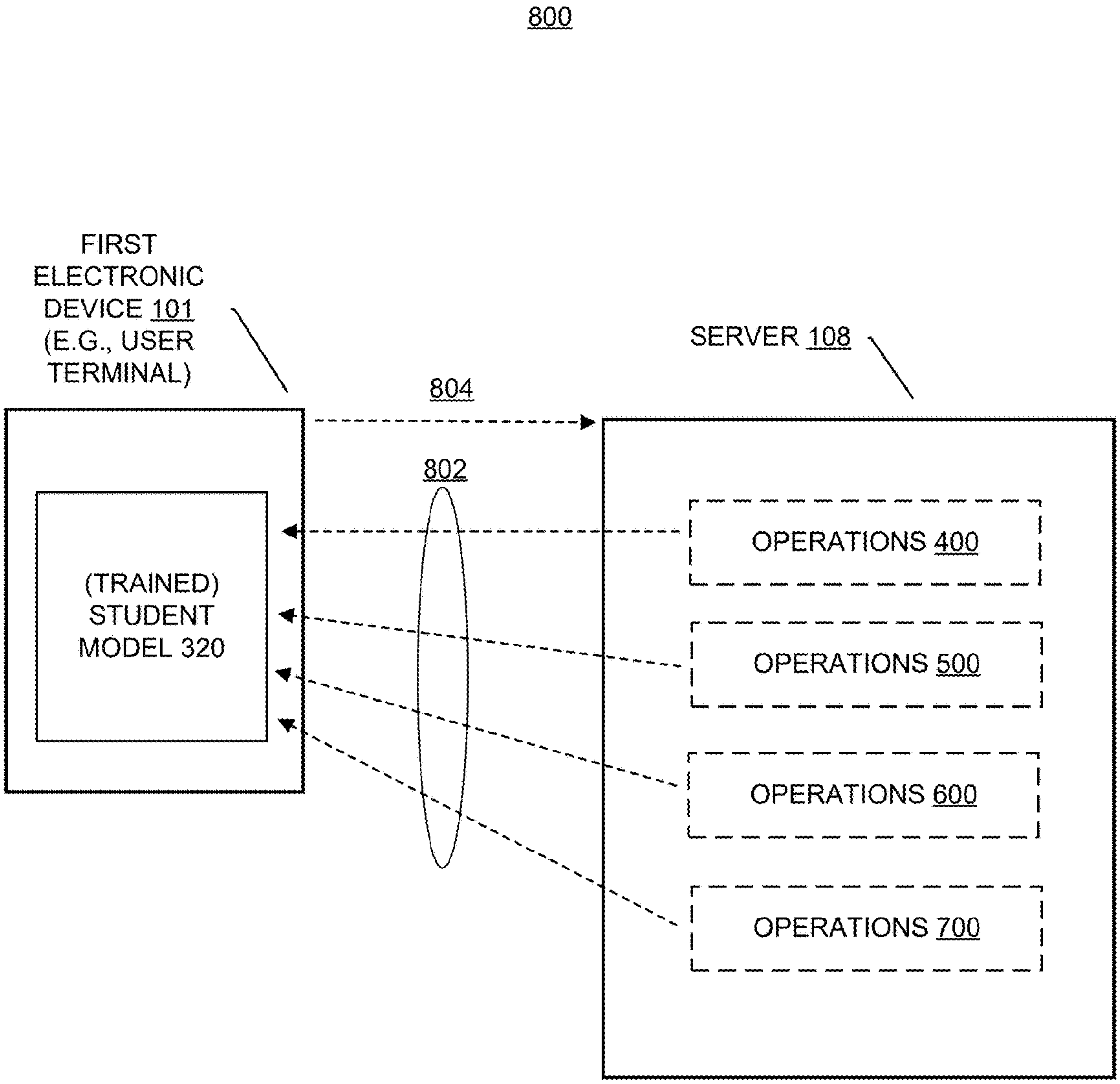

800
FIRST ELECTRONIC DEVICE 101 (E.G., USER TERMINAL)
SERVER 108
804
802
(TRAINED) STUDENT MODEL 320
OPERATIONS 400
OPERATIONS 500
OPERATIONS 600
OPERATIONS 700

FIG. 8

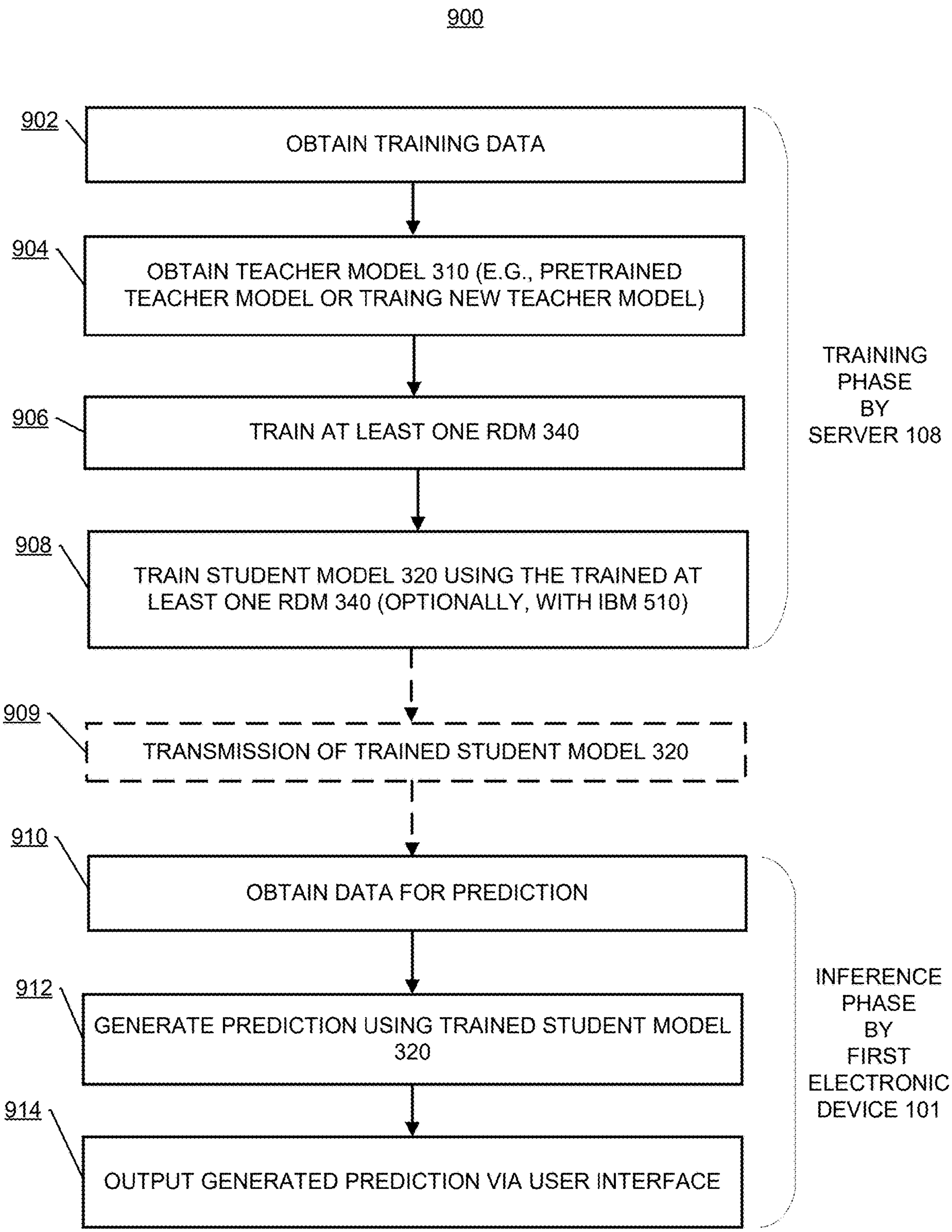

900
902
OBTAIN TRAINING DATA
904
OBTAIN TEACHER MODEL 310 (E.G., PRETRAINED TEACHER MODEL OR TRAING NEW TEACHER MODEL)
906
TRAIN AT LEAST ONE RDM 340
908
TRAIN STUDENT MODEL 320 USING THE TRAINED AT LEAST ONE RDM 340 (OPTIONALLY, WITH IBM 510)
TRAINING PHASE BY SERVER 108
909
TRANSMISSION OF TRAINED STUDENT MODEL 320
910
OBTAIN DATA FOR PREDICTION
912
GENERATE PREDICTION USING TRAINED STUDENT MODEL 320
914
OUTPUT GENERATED PREDICTION VIA USER INTERFACE
INFERENCE PHASE BY FIRST ELECTRONIC DEVICE 101

FIG. 9

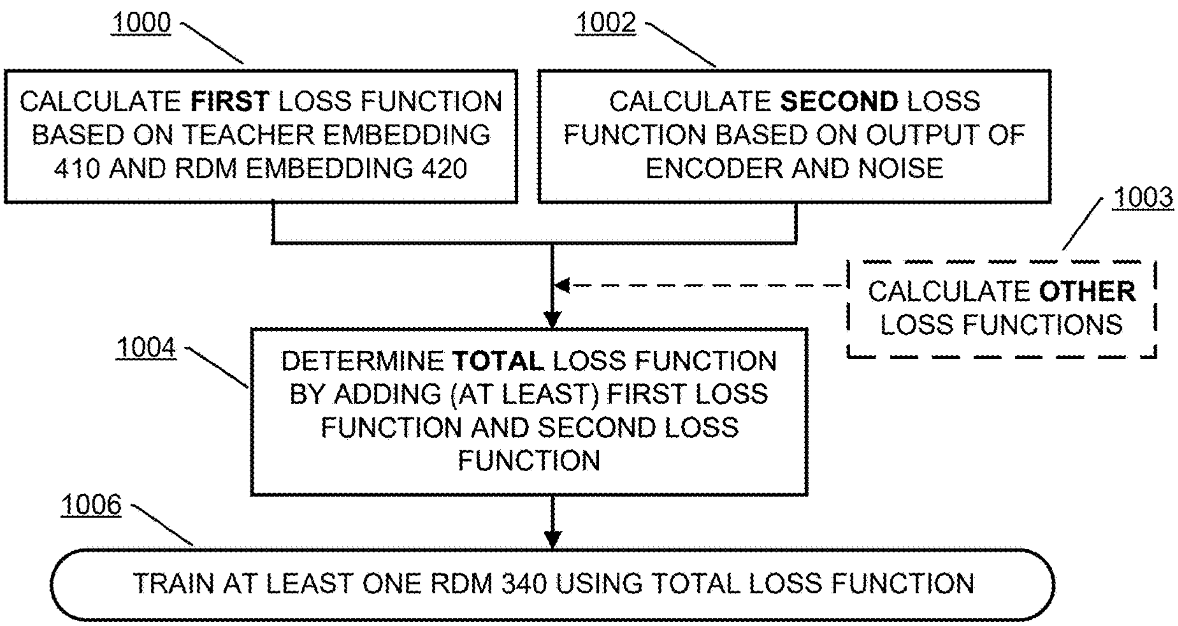

1000
CALCULATE FIRST LOSS FUNCTION BASED ON TEACHER EMBEDDING 410 AND RDM EMBEDDING 420
1002
CALCULATE SECOND LOSS FUNCTION BASED ON OUTPUT OF ENCODER AND NOISE
1003
CALCULATE OTHER LOSS FUNCTIONS
1004
DETERMINE TOTAL LOSS FUNCTION BY ADDING (AT LEAST) FIRST LOSS FUNCTION AND SECOND LOSS FUNCTION
1006
TRAIN AT LEAST ONE RDM 340 USING TOTAL LOSS FUNCTION

FIG. 10

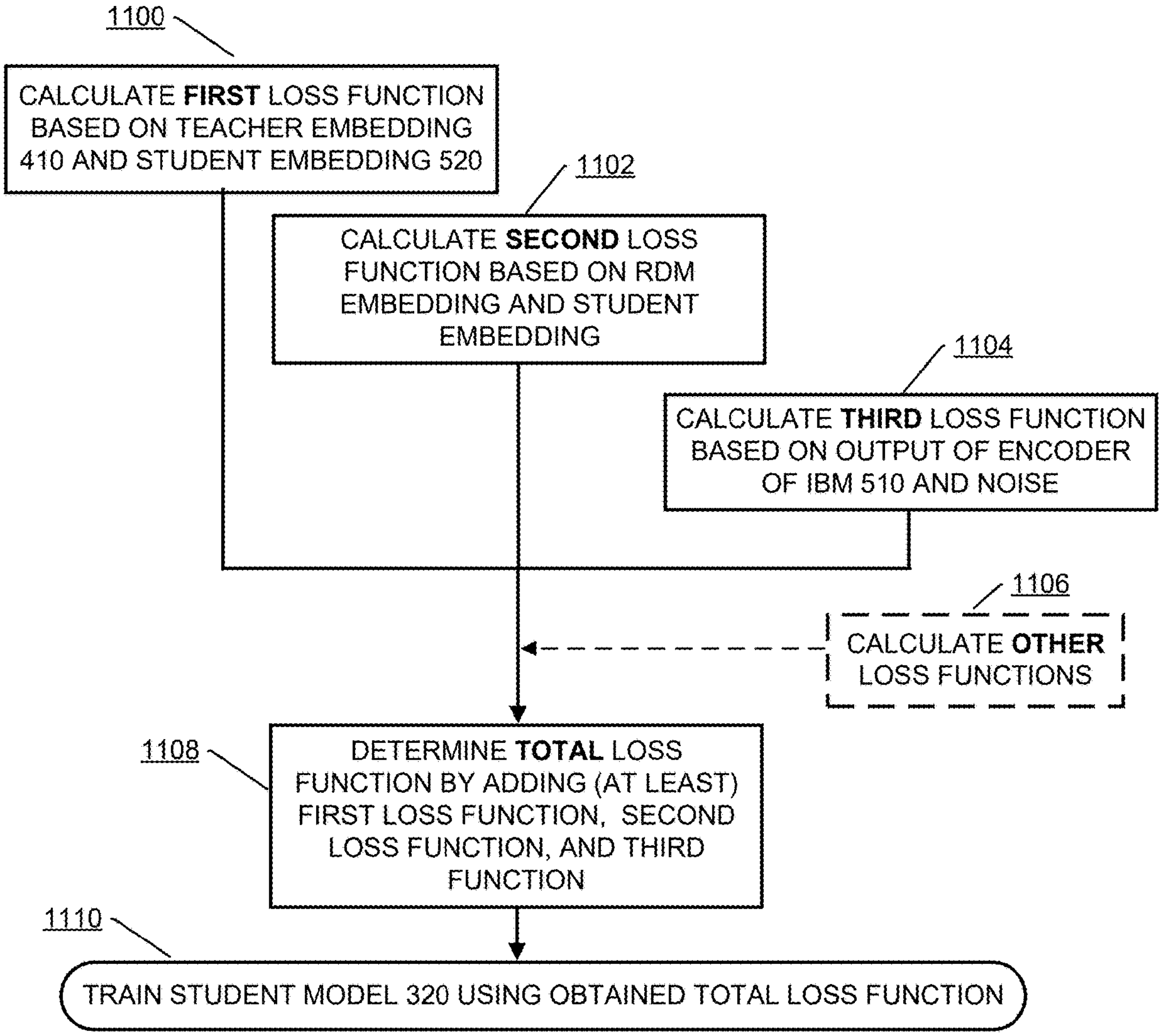

1100
CALCULATE FIRST LOSS FUNCTION BASED ON TEACHER EMBEDDING 410 AND STUDENT EMBEDDING 520
1102
CALCULATE SECOND LOSS FUNCTION BASED ON RDM EMBEDDING AND STUDENT EMBEDDING
1104
CALCULATE THIRD LOSS FUNCTION BASED ON OUTPUT OF ENCODER OF IBM 510 AND NOISE
1106
CALCULATE OTHER LOSS FUNCTIONS
1108
DETERMINE TOTAL LOSS FUNCTION BY ADDING (AT LEAST) FIRST LOSS FUNCTION, SECOND LOSS FUNCTION, AND THIRD FUNCTION
1110
TRAIN STUDENT MODEL 320 USING OBTAINED TOTAL LOSS FUNCTION

FIG. 11

CONTROLLED INFORMATION FLOW FROM TEACHER MODEL TO STUDENT MODEL FOR KNOWLEDGE DISTILLATION AND TRANSFER

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is based on and claims priority under 35 U.S.C. § 119 to U.S. Provisional Patent Application No. 63/558,424, filed on Feb. 27, 2024, in the United States Patent and Trademark Office, the disclosure of which is incorporated by reference herein its entirety.

BACKGROUND

1. Field

[0002] The disclosure relates to a system and a method for facilitating a transfer of knowledge from a larger model (a teacher model) to a smaller model (a student model), for example, in the fields of machine learning of artificial intelligence.

2. Description of Related Art

[0003] Deep neural networks are ubiquitous in their usage today. While large models are constantly pushing the limits of what is technically possible, their practical usage is limited. Large models require huge computation resources to train but most importantly even for deployment (i.e., inference). In the deployment phase, the model is not being trained but used as a tool. However, given the ever-increasing size, run-time memory, and computation requirements, most enterprises provide cloud-based solutions to their clients (i.e., users). This allows the clients to use large models without having to worry about the computational burden. However, this opens up to multiple issues such as constant need for an internet connectivity, bandwidth requirements, load managing at a server, and client data privacy issues.

[0004] One possible solution is to use a smaller model that fits on resource and computation-constrained devices. Knowledge Distillation (KD) is a method of improving the performance of a "student model" (a smaller model) by using a "teacher model" (a larger model) to teach insights of the teacher model to the student model. The teacher model, on account of larger modeling capacity, is capable of learning complex relationships in the data. Thus, the teacher model is better at performing target tasks than the student model.

[0005] KD attempts to help a training of the student model by showing the student model the insights learned by the teacher model by transferring so-called "dark knowledge" in the logits of a teacher classifier to a student classifier. To facilitate a transfer of the dark knowledge, Kullback-Leibler (KL) divergence between a softened version of a predicted probability distribution of the teacher model and the student model is used, in addition to a classical supervised training of the student model. The dark knowledge is one way of quantifying the insights learned by the teacher model. In general, how to quantify the insights from the teacher model and how to transfer the insights to the student model remain open questions.

[0006] In the related art, to facilitate a transfer of knowledge from the teacher model to the student model, training an intermediate model, so-called 'Teacher Assistant' (TA) model, has been proposed. However, training the intermediate model (the TA model) from scratch has a knowledge transfer problem itself, and thus, is computationally expensive. Further, the intermediate model is larger than the student model, and thus, is more expensive to train. Thus, this is a need for a solution that is less expensive than the training of the intermediate model (the TA model) of the related art.

SUMMARY

[0007] The disclosure is directed to a system and a method for facilitating a transfer of knowledge from a teacher model to a student model, which are less expensive than a system and a method using an intermediate model (a teaching assistant (TA) model) for the transfer of knowledge from the teacher model to the student model.

[0008] The disclosure is directed to a system and a method using a rate-distortion module (RDM) trained to mimic teacher assistants used for knowledge distillation (KD) in a field of artificial intelligence (AI).

[0009] The disclosure is directed to a system and a method for training the RDM to mimic a plurality of TA models.

[0010] The disclosure is directed to a system and a method for using an information bottleneck module (IBM) to regularize a training of the student model during the transfer of knowledge.

[0011] According to an aspect of the disclosure, a computer-implemented method performed by an electronic device configured to distill knowledge from a teacher model and transfer the distilled knowledge to a student model, includes: obtaining training data; obtaining the teacher model; training at least one rate distortion module (RDM) that is not a teacher assistant (TA) model; training the student model using the trained at least one RDM; and transmitting the trained student model to a first electronic device.

[0012] In an embodiment, the transmitting the trained student model to a first electronic device, includes: receiving, via a user interface of a display in the electronic device, a user's first command to select the trained student model, receiving, via the user interface, the user's second command to transmit the trained student model to the first electronic device, and transmitting, based on the user's second command, the trained student model to a first electronic device.

[0013] In an embodiment, the teacher model is a large language model (LLM) and the student model is a small language model (SLM).

[0014] In an embodiment, the teacher model is a pretrained model that is pretrained by an external device.

[0015] In an embodiment, the teacher model is a large language model trained by the electronic device.

[0016] In an embodiment, the first electronic device is a user terminal.

[0017] In an embodiment, the training the at least one RDM, includes: calculating a first loss function based on teacher embeddings and RDM embeddings; calculating a second loss function based on an output of an encoder and noise; determining a total loss function by adding at least the first loss function and the second loss function; and training the at least one RDM using the determined total loss function.

[0018] In an embodiment, the computer-implemented method further includes calculating other loss functions, wherein the determining the total loss function by adding at

least the first loss function and the second loss function comprises determining the total loss function by the first loss function, the second loss function, and the other loss functions.

[0019] In an embodiment, the training the student model using the trained at least one RDM, includes: calculating a first loss function based on teacher embeddings and student embeddings; calculating a second loss function based on RDM embeddings and the student embeddings; calculating a third loss function based on an output of an encoder of an information bottleneck module (IBM) and noise; determining a total loss function by adding at least the first loss function, the second loss function, and the third loss function; and training the at least one RDM using the determined total loss function.

[0020] In an embodiment, the computer-implemented method further includes calculating other loss functions, wherein the determining the total loss function by adding at least the first loss function, the second loss function, and the third loss function comprises determining the total loss function by the first loss function, the second loss function, the third loss function, and the other loss functions.

[0021] According to an aspect of the disclosure, an electronic device configured to distill knowledge from a teacher model and transfer the distilled knowledge to a student model, includes: at least one memory; a display displaying a user interface; and at least one processor operatively connected with the at least one memory and the display; wherein the at least one processor is configured to perform: obtaining training data; obtaining the teacher model; training at least one rate distortion module (RDM) that is not a teacher assistant (TA) model; training the student model using the trained at least one RDM; and transmitting the trained student model to a first electronic device.

[0022] In an embodiment, the at least one processor is further configured to perform: receiving, via the user interface, a user's first command to select the trained student model, receiving, via the user interface, the user's second command to transmit the trained student model to the first electronic device, and transmitting, based on the user's second command, the trained student model to a first electronic device.

[0023] In an embodiment, the teacher model is a large language model (LLM) and the student model is a small language model (SLM).

[0024] In an embodiment, the teacher model is a pretrained model that is pretrained by an external device.

[0025] In an embodiment, the teacher model is a large language model trained by the electronic device.

[0026] In an embodiment, the first electronic device is a user terminal.

[0027] In an embodiment, the at least one processor is further configured to perform: calculating a first loss function based on teacher embeddings and RDM embeddings; calculating a second loss function based on an output of an encoder and noise; determining a total loss function by adding at least the first loss function and the second loss function; and training the at least one RDM using the determined total loss function.

[0028] In an embodiment, the at least one processor is further configured to perform: calculating other loss functions, and determining the total loss function by the first loss function, the second loss function, and the other loss functions.

[0029] In an embodiment, the at least one processor is further configured to perform: calculating a first loss function based on teacher embeddings and student embeddings; calculating a second loss function based on RDM embeddings and the student embeddings; calculating a third loss function based on an output of an encoder of an information bottleneck module (IBM) and noise; determining a total loss function by adding at least the first loss function, the second loss function, and the third loss function; and training the at least one RDM using the determined total loss function.

[0030] In an embodiment, the at least one processor is further configured to perform: calculating other loss functions, and determining the total loss function by the first loss function, the second loss function, the third loss function, and the other loss functions.

BRIEF DESCRIPTION OF THE DRAWINGS

[0031] The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0032] FIG. **1** illustrates example components of an electronic device in accordance with some embodiments of the disclosure;

[0033] FIG. **2** illustrates an example block diagram of a server in accordance with some embodiments of the disclosure;

[0034] FIG. **3** illustrates a set of operations of controlled information flow for knowledge distillation (KD) in accordance with some embodiments of the disclosure;

[0035] FIG. **4** illustrates a set of operations of controlled information flow for KD in accordance with some embodiments of the disclosure;

[0036] FIG. **5** illustrates a set of operations of controlled information flow for KD in accordance with some embodiments of the disclosure;

[0037] FIG. **6** illustrates a set of operations of controlled information flow for KD in accordance with some embodiments of the disclosure;

[0038] FIG. **7** illustrates a set of operations of controlled information flow for KD in accordance with some embodiments of the disclosure;

[0039] FIG. **8** illustrates examples of practical applications of the above operations in accordance with some embodiments of the disclosure;

[0040] FIG. **9** illustrates example set of operations in accordance with some embodiments of the disclosure;

[0041] FIG. **10** illustrates example set of operations in accordance with some embodiments of the disclosure; and

[0042] FIG. **11** illustrates example set of operations in accordance with some embodiments of the disclosure.

DETAILED DESCRIPTION

[0043] The terms as used in the disclosure are provided to merely describe specific embodiments, not intended to limit the scope of other embodiments. Singular forms include plural referents unless the context clearly dictates otherwise. The terms and words as used herein, including technical or scientific terms, may have the same meanings as generally understood by those skilled in the art. The terms as generally defined in dictionaries may be interpreted as having the same or similar meanings as or to contextual meanings of the relevant art. Unless otherwise defined, the terms should not be interpreted as ideally or excessively formal meanings. Even though a term is defined in the disclosure, the term should not be interpreted as excluding embodiments of the disclosure under circumstances.

[0044] The blocks in each flowchart and combinations of the flowcharts may be performed by one or more computer programs which include computer-executable instructions. The entirety of the one or more computer programs may be stored in a single memory or the one or more computer programs may be divided with different portions stored in different multiple memories.

[0045] Any of the functions or operations described herein may be processed by one processor or a combination of processors. The one processor or the combination of processors is circuitry performing processing and includes circuitry like an application processor (AP), a communication processor (CP), a graphical processing unit (GPU), a neural processing unit (NPU), a microprocessor unit (MPU), a system on chip (SoC), an IC, or the like.

[0046] The disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as "A or B", "at least one of A and B", "at least one of A or B", "A, B, or C", "at least one of A, B, and C", and "at least one of A, B, or C", may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as "1st" and "2nd", or "first" and "second" may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term "operatively" or "communicatively", as "coupled with", "coupled to", "connected with", or "connected to" another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., via a wire), wirelessly, or via a third element.

[0047] A "unit" or a "module" used in the disclosure refer to a hardware component such as a processor or a circuit, and/or a software component executed by a hardware component such as a processor. The "unit" or the "module" may be implemented by a program that is stored in a storage medium which may be addressed, and is executed by a processor. For example, a "unit", "module" may be implemented by components such as software components, object-oriented software components, class components, and task components, processes, functions, attributes, procedures, sub-routines, segments of a program code, drivers, firmware, a micro code, a circuit, data, a database, data structures, tables, arrays and parameters.

[0048] FIG. **1** illustrates example components of the electronic device in accordance with some embodiments of the disclosure.

[0049] In FIG. **1**, a (first) electronic device **101** may communicate with a second electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or a third electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). In one embodiment, the (first) electronic device **101** may communicate with the third electronic device **104** via the server **108**. Throughout the disclosure, the first electronic device **101** may be referred to as 'the electronic device **101**.' Hereinafter, components of the electronic device **101** are described. Those components of the electronic device **101** may be also included in the second electronic device **102** or the third electronic device **104**. The first electronic device **101**, the second electronic device **102**, or the third electronic device **104** may be configured to perform methods, steps, or operations described in the disclosure.

[0050] In an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input device **150**, a sound output circuit **155**, a display **160**, an audio circuit **170**, a sensor **176**, an interface **177**, a connection terminal **178**, a haptic circuit **179**, a camera **180**, a power management circuit **188**, a battery **189**, a communication circuit **190**, or an antenna **197**.

[0051] In an embodiment, at least one (e.g., the display **160**, the sensor **176**, or the camera **180**) of the components may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In an embodiment, some of the components may be implemented as single integrated circuitry. For example, the sensor **176** (e.g., a fingerprint sensor, an iris sensor, or an illuminance sensor) may be implemented as embedded in the display **160** (e.g., a touch screen). In an embodiment, the electronic device **101** may be a user equipment, a user terminal, a smartphone, a tablet personal computer (PC), a laptop, a PC and/or a server.

[0052] In an embodiment, the at least one processor **120** (or the main processor **121** or the auxiliary processor **123**) may be implemented in hardware, firmware, or a combination of hardware and software. The at least one processor **120** (or the main processor **121** or the auxiliary processor **123**) may include one or more of a central processing unit (CPU), a graphics processing unit (GPU), an accelerated processing unit (APU), a many integrated core (MIC), a field-programmable gate array (FPGA), a digital signal processor (DSP), a neural processing unit (NPU), a hardware accelerator, or a machine learning accelerator. The at least one processor **120** (or the main processor **121** or the auxiliary processor **123**) are able to perform control of any one or any combination of the other components of the computing device, and/or perform an operation or data processing relating to communication. The at least one processor **120** (or the main processor **121** or the auxiliary processor **123**) execute one or more programs stored in a memory.

[0053] The at least one processor **120** (or the main processor **121** or the auxiliary processor **123**) may be implemented as one or more multi-core processors that include one or more cores (e.g., homogeneous multi-cores or heterogeneous multi-cores). When a plurality of cores are included in the at least one processor **120** (or the main processor **121** or the auxiliary processor **123**), each of the cores includes a cache memory, and a common cache shared by the cores may be included in the at least one processor **120** (or the main processor **121** or the auxiliary processor **123**). Each of the cores may independently read and execute program instructions or each of the cores may read and execute one or more portions of program instructions.

[0054] In an embodiment, the at least one processor **120** (or the main processor **121** or the auxiliary processor **123**) may refer to a system-on-a-chip (SoC) in which one or more cores and other electronic components are integrated, a single core processor, a multicore processor, or a core included in the single core processor or the multicore processor, wherein the core may be implemented as a CPU, a GPU, an APU, an MIC, a FPGA, a DSP, an NPU, a hardware accelerator, or a machine learning accelerator, but the embodiments of the disclosure are not limited thereto.

[0055] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. In one embodiment, as at least part of the data processing or computation, the processor **120** may load a command or data received from another component (e.g., the sensor **176** or the communication circuit **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**.

[0056] In one embodiment, the processor **120** may include a main processor **121** (e.g., a CPU or an AP), and an auxiliary processor **123** (e.g., a graphics processing unit (GPU), an image signal processor (ISP), a sensor hub processor, or a CP) that is operable independently from, or in conjunction with, the main processor **121**. Additionally, or alternatively, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The processor **120** may refer to or correspond to one or more processors. For example, the electronic device **101** may include two or more processors like the processor **120**. In an embodiment, the main processor **121** and the auxiliary processor **123** may comprise processing circuitry.

[0057] The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**. The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display **160**, the sensor **176**, or the communication circuit **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). In one embodiment, the auxiliary processor **123** (e.g., an ISP or a CP) may be implemented as part of another component (e.g., the camera **180** or the communication circuit **190**) functionally related to the auxiliary processor **123**.

[0058] For example, the processor **120** of the electronic device **101** may invoke at least one of the one or more instructions stored in the memory **130**, and execute the at least one of the one or more instructions, with or without using one or more other components under the control of the processor **120**. This allows the electronic device **101** to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The memory **130**, which may be a machine-readable storage medium, may be provided in the form of a non-transitory storage medium. Wherein, the term "non-transitory" simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the memory **130** (the storage medium) and where the data is temporarily stored in the memory **130**. In an embodiment, the electronic device **101** may comprise one or more processors (e.g., the main processor **121** and the auxiliary processor **123**), and the one or more instructions may be executed by the one or more processors individually or collectively, thereby causing the electronic device **101** to perform any combination of one or more operations (or functions, steps) described herein.

[0059] In an embodiment, the memory **130** may include a random-access memory (RAM), a read only memory (ROM), and/or another type of dynamic or static storage device (e.g., a flash memory, a magnetic memory, and/or an optical memory) that stores information and/or instructions for use by the processor **120**. In an embodiment, the memory **130** may contain information and/or software related to the operation and use of the electronic device **101**. For example, the memory **130** may include a hard disk (e.g., a magnetic disk, an optical disk, a magneto-optic disk, and/or a solid-state disk), a compact disc (CD), a digital versatile disc (DVD), a floppy disk, a cartridge, a magnetic tape, or another type of non-transitory computer-readable medium, along with a corresponding drive.

[0060] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**. The non-volatile memory **134** may include the internal memory **136** or external memory **138**. The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

[0061] One or more embodiments of the disclosure may be implemented as software (e.g., the operating system **142**, the application **146**, the middleware **144**) including one or more instructions that are stored in the memory **130** (comprising one or more storage medium) that is readable by the electronic device **101**.

[0062] In an embodiment, the input device **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user, the second electronic device **102**, or the third electronic device **104**) of the electronic device **101**. The input device **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

[0063] In an embodiment, the sound output circuit **155** may output sound signals to the outside of the electronic device **101**. The sound output circuit **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing recorded data. The receiver may be used for receiving incoming calls. According to some embodiments, the receiver may be implemented as separate from, or as part of the speaker.

[0064] In an embodiment, the display **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display **160** may include, for example, a display device, a hologram device, or a projector and control circuitry to control a corresponding one of the display device, hologram device, and projector. According to some embodiments, the display **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

[0065] In an embodiment, the audio circuit **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio circuit **170** may obtain the sound via the input device **150** or output the sound via the sound output circuit **155** or a headphone of an external electronic device (e.g., the second electronic device **102** or the third electronic device **104**) directly (e.g., via a wire) or wirelessly coupled with the electronic device **101**.

[0066] In an embodiment, a sensor **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state.

[0067] In an embodiment, the interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external entity (e.g., the second electronic device **102**, the third electronic device **104**, or the server **108**) directly (e.g., via a wire) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high-definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0068] In an embodiment, the connection terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the second electronic device **102**, the third electronic device **104**, or the server **108**). According to some embodiments, the connection terminal **178** may include, for example, an HDMI connector, a USB connector, an SD card connector, or an audio connector (e.g., a headphone connector).

[0069] In an embodiment, the haptic circuit **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic circuit **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0070] In an embodiment, the camera **180** may capture a still image or moving images (or a set or one or more still images, or video data). According to some embodiments, the camera **180** may include one or more lenses, image sensors, ISPs, or flashes.

[0071] In an embodiment, the power management circuit **188** may manage power supplied to the electronic device **101**. According to some embodiments, the power management circuit **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0072] In an embodiment, the battery **189** may supply power to at least one component of the electronic device **101**. According to some embodiments, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0073] In an embodiment, the communication circuit **190** may include a transceiver-like component (e.g., a transceiver and/or a separate receiver and transmitter) that enables the electronic device **101** to communicate with other devices (e.g., the second electronic device **102**, the third electronic device **104**, or the server **108**), such as via a wired connection, a wireless connection, or a combination of wired and wireless connections. The communication circuit **190** may permit the electronic device **101** to receive information from another device and/or provide information to another device. For example, the communication circuit **190** may include an Ethernet interface, an optical interface, a coaxial interface, an infrared interface, a radio frequency (RF) interface, a universal serial bus (USB) interface, a Wi-Fi interface, a cellular network interface, or the like. In an embodiment, the communication circuit **190** may be a communication 'interface' used to connect the electronic device **101** with the other devices.

[0074] In an embodiment, the communication circuit **190** may include one or more CPs that are operable independently from the processor **120** (e.g., an AP) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication circuit **190** may include a wireless communication circuit **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication circuit **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module).

[0075] A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, Wi-Fi direct, or IR data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN)). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication circuit **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)).

[0076] The antenna **197** may transmit or receive a signal or power to or from the outside (e.g., an external electronic device) of the electronic device **101**. According to an embodiment, the antenna **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna **197** may include a plurality of antennas (e.g., array antennas).

[0077] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) between or among the components via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0078] In an embodiment, a set of components (e.g., one or more components) of the electronic device **101** may perform one or more functions described as being performed by another set of components of the electronic device **101**.

[0079] FIG. **2** is an example block diagram of the server **108** in accordance with some embodiments of the disclosure. In FIG. **2**, the server **108** may be an electronic device including a communication interface **202**, a processor **204**, and a memory **206**. However, the components of the server **108** are not limited to the above-described examples. The server **108** may include more or fewer components than the aforementioned components. According to an embodiment of the disclosure, some or all of the communication interface **202**, the processor **204**, and the memory **206** may be implemented as a single chip, and the processor **204** may include one or more processors.

[0080] The communication interface **202** is a component for transmitting and receiving signals (e.g., control commands and data) with an external device (e.g., the electronic device **101**) by wire or wirelessly, and may be configured to include a communication chipset that supports various communication protocols. The communication interface **202** may receive a signal from an external source and output the signal to the processor **204**, or may transmit a signal output by the processor **204** to an external source. According to an embodiment of the disclosure, referring to FIG. **1**, the communication interface **11** may perform communication with the electronic device **101** through the second network **199**.

[0081] In an embodiment, the server **108** may generate a language model and transfer the language model to the electronic device **101** via the communication interface **202**.

[0082] The processor **204** is configured to control a series of processes so that the server **108** performs any combination of operations in accordance with embodiments described above. The processor **204** may include one or a plurality of processors, each comprising processing circuitry. The one or plurality of processors may be a general-purpose processor such as a CPU, an AP, a DSP; a graphics-only processor such as a GPU or a vision processing unit (VPU); or an Artificial Intelligence (AI)-only processor such as an NPU.

[0083] For example, when the one or plurality of processors are AI-only processors, the AI-only processors may be designed in a hardware structure specialized for processing a specific AI model. In an embodiment, the processor **204** may be implemented in a manner similar to that of the processor **120** in FIG. **1**.

[0084] The processor **204** may write data to the memory **206** or read data stored in the memory **206**, and, in particular, may execute one or more programs or one or more instructions stored in the memory **206** to process data according to a predefined operation rule or an AI model. Accordingly, the processor **204** may perform any combination of operations described in the above embodiments. Any combination of operations described as being performed by the server **108** in the above embodiments may be considered as being performed by the processor **204** unless otherwise specified.

[0085] According to an embodiment of the disclosure, the memory **206** may store one or more instructions associated with operations of the language model.

[0086] In an embodiment, the server **108** may comprises a plurality of processors, and the one or more instructions stored in the memory **206** may be executed by the plurality of processors individually or collectively, causing the server **108** to perform any combination of operations described above.

[0087] The memory **206**, which is a component for storing various programs or data, may be composed of one or more storage medium, such as ROM, RAM, hard disks, CD-ROM, and DVDs, a floppy disk, a cartridge, a magnetic tape, or another type of non-transitory computer-readable medium, along with a corresponding drive. The memory **206** may not exist separately but may be included in the processor **204**. The memory **206** may be implemented as a volatile memory, a non-volatile memory, or a combination of a volatile memory and a non-volatile memory. In an embodiment, the memory **206** may be implemented in a manner similar to that of the memory **130** in FIG. **1**.

[0088] The memory **206** may store one or more programs, program codes, or instructions for performing any combinations of operations according to embodiments of the disclosure describe above. The memory **206** may provide stored data to the processor **204**, in response to a request by the processor **204**.

[0089] The server **108** may include a display **208** that is configured to show a user interface **210**. A user of the server **108** may enter a command, through the user interface **210**, to instruct the server **108** to perform functions.

[0090] According to an embodiment of the disclosure, the server **108** may perform computer-implemented methods of the disclosure, which are described below.

[0091] The system of the disclosure leverages complex data insights learned by the teacher model and transfers the complex data insights to the student model. The teacher model can be obtained as a pretrained model or trained from scratch before the knowledge transfer and is frozen during the entirety of the knowledge transfer. In some embodiments, both of the teacher model and the student model are neural networks.

[0092] In some embodiments, the system of the disclosure includes the two phrases: a training phase and an inference phase. In some embodiments, the training phase has two stages: stage 1 and stage 2.

Stage 1 of the Training Phase

[0093] In the stage 1 of the training phase, at least one rate distortion module (RDM) is trained by at least one processor configured to process teacher embeddings (obtained from the teacher model). The at least one RDM is a lightweight model that controls the amount of information passing through it. By controlling the amount of information, different RDMs may mimic different Teacher Assistants (TAs), or different RDMs may be used in lieu of different TAs. This use of the RDM may provide the student model with different granularities of information during the stage 2 of the training phase, which is described below.

[0094] In some embodiments, the above training of the at least one RDM may be conducted by the processor of the server **108**. In some embodiments, the at least one RDM may be stored or saved in the memory of the server **108**.

Stage 2 of the Training Phase

[0095] Once the stage 1 of the training phase is completed, the at least one RDM is frozen (i.e., no longer being trained). During this stage 2, the embedding predictions from the teacher model and the at least one RDM are used to train the student model. In some embodiments, optionally, during this stage 2, an information bottleneck module (IBM), which is described below, may be connected to the student model and be used to regularize embeddings of the student model.

Inference Phase

[0096] After the student model is trained during the stage 2, the trained student model can be used for multiple applications such as image classification, audio classification, zero-shot classification, zero-shot retrievals, provide modality specific encoders for multimodal large language models or distill large language models themselves.

[0097] In some embodiments, the student model is trained (through the stage 1 and 2) in a first electronic device (e.g., the server **108**X) and the trained student model may be transferred to a second electronic device (e.g., the first electronic device **101** or a mobile device such as a smartphone, a tablet, or a portable laptop). In some embodiments, the second electronic device receives or obtains data (e.g., images) from a data source and use the trained student model to obtain predictions about the data.

[0098] Throughout the disclosure, the term 'module' (e.g., as in the at least one RDM and the IBM) indicates or correspond to hardware only (e.g., the processor **120**), software only (e.g., computer codes or programs stored in the memory **130**), or combinations of the hardware and the software. For example, the at least one RDM and the IBM may be implemented by computer codes (e.g., Python or any other programming language).

[0099] The at least one RDM (one or more RDMs) and the IBM may be combined differently for different embodiments of the disclosure. Some example combinations of the at least one RDM and the IBM ((Stage 1, Stage 2)) are introduced herein: (one RDM in the stage 1, with the IBM in the stage 2); (one RDM in the stage 1, without the IBM in the stage 2); (multiple RDMs in the stage 1, with the IBM in the stage 2); (multiple RDMs in the stage 1, without the IBM in the stage 2); and (no RDM in the stage 1, the IBM in the stage 2).

[0100] FIG. **3** illustrates operations **300** of controlled information flow for KD in accordance with some embodiments of the disclosure. For example, the operations **300** of controlled information flow for KD may be stored in the memory **130** as computer codes and may be executed by the processor **120**. For example, the operations **300** of controlled information flow for KD may be implemented with hardware only (e.g., the processor **120**), software only (e.g., computer codes or programs stored in the memory **130**), or combinations of the hardware and the software.

[0101] As shown in FIG. **3**, the operations **300** of controlled information flow for KD includes the teacher model **310**, the student model **320**, and a plurality of RDMs **340** (or at least one RDM **340**). As shown in FIG. **3**, the RDM **340** includes an encoder **344** and a decoder **348**. The encoder **344** receives input embeddings **342**. The decoder **348** receives an input from the encoder **344** and noise **346**. The decoder **348** outputs reconstruction embeddings **349**. As shown in of FIG. **3**, the operations **300** of controlled information flow for KD, the RDMs **340** use the teacher model's embeddings to generate embeddings that mimic Teaching Assistants (TAs). The RDMs **340** process the teacher model's embeddings through a rate constrained communication channel.

[0102] The RDM **340** is further described below.

[0103] As an example, as illustrated in FIG. **3**, the RDM **340** includes only three layers, which are significantly cheaper in computation than the TAs. In other words, the RDM **340** may need about a quarter of computations compared with the TAs, which correspond to lower training cost. Also, a knowledge distillation system having the RDM **340** may have classification accuracy that is similar to the classification accuracy of a knowledge distillation system having the TAs.

[0104] Although FIG. **3** illustrates that the RDM **340** has three layers, the disclosure is not limited to the above embodiment. In some embodiments, the RDM **340** may have more than three layers. In some embodiments, the RDM **340** may have less than three layers.

[0105] Compared to the TAs, the RDMs **340** are much cheaper to train and use because the RDMs process the teacher model's embeddings and the RDMs **340** do not learn low level feature extractors for an input. By varying the rate constraint, the RDMs **340** may be capable of simulating different TAs.

[0106] The operations **300** of controlled information flow for KD distill knowledge (or the insights) from the teacher model **310** to the student model **320**, without using intermediate TA models. In operations **300** of controlled information flow for KD, the teacher model's penultimate layer embeddings are processed by the RDMs **340** that imposes a constraint on the amount of information through the RDMs **340**.

[0107] Using the RDM **340** for the KD has the following advantages. Based on different rate constraints, the RDM **340** may mimic different TAs. In one embodiment, the RDM **340** may include fewer than three (3) hidden layers. Because the RDM **340** operates on the embeddings of the teacher model **310**, the RDM **340** may not learn any low-level feature extractors. Accordingly, the RDM **340** may be cheaper to train than the TA, and an inference on the RDM **340** is cheaper than the TA. Thus, the computational burden of the RDM **340** during the training of the student model **320** is reduced.

[0108] The operations of the RDM **340** are described below.

[0109] The RDM **340** is configured to compress a signal (under rate-distortion theory). Given an input X (e.g., in FIG. **3**, the input embeddings **342** or the image embeddings), the RDM **340** is operated to find a mapping from the input X to its compressed version $\hat{X}$ such that $\hat{X}$ has minimal information about X. But, at the same time, the distortion of the input does not exceed D_0. An optimization problem of the RDM **340** is expressed as the following (1) expression:

$$\min_{X \to \hat{X}: D(X;\hat{X}) \le D_0} I(X;\hat{X}) \tag{1}$$

[0110] Here, $D(\cdot,\cdot)$ is distortion measure and $I(\cdot,\cdot)$ denotes mutual information between two random variables. The above (1) expression is converted to an unconstrained optimization objective of the form, as shown below:

$$\min R \cdot D(X;\hat{X}) + I(X;\hat{X}) \tag{2}$$

[0111] Here, R determines the trade-off between information rate and distortion. A larger R corresponds to more emphasis on minimizing the distortion at the cost of a higher information rate and vice-versa. The above (2) expression may be understood in the context of lossy compression that compresses as much as possible while allowing tolerable distortion.

[0112] The RDM **340** is operable under the rate-distortion theory that is applicable to compression and also to problems like joint source-channel coding where the compressed representation is subject to noise. As shown in FIG. **3**, the encoded representation of the input X is denoted as Y and its noisy version as $\hat{Y}$. The independent noise added to encoded representation is denoted as Z.

[0113] The rate-distortion theory indicates what an optimization problem to be solved, but the rate-distortion theory may not provide how to solve the optimization problem. When the encoder **344** and the decoder **348** in FIG. **3** are neural networks, the optimization problem is compounded because the objective may not be computed ($I(X;\hat{X})$ is intractable).

[0114] Thus, the operations **300** of controlled information flow for KD may use variational approximations to compute an upper bound on the objective, which, in turn, will perform gradient descent to learn the encoder **344** and the decoder **348**. $q(\hat{Y})$ is denoted as an approximation of true but unknown distribution of $\hat{Y}$, $p(\hat{Y})$. Then, an upper-bound on $I(X;\hat{X})$ may be computed as shown below:

$$I(X;\hat{X}) \leq I(Y;\hat{Y}) \leq H_q(\hat{Y}) + H(\hat{Y} \mid Y) \tag{3}$$

[0115] Here, $H(\cdot)$ denotes the entropy, and $H_q(\hat{Y})$ denotes the cross entropy computed using the distribution $q(\hat{Y})$. The first (left) inequality of the above expression (3) follows from the data processing inequality. The second inequality (right) of the above expression (3) follows because cross-entropy may be greater than entropy. Finally, note that $H(\hat{Y}|Y)$ is constant because the noise is independent of the encoder parameter and the decoder parameter.

[0116] In the above expression (3), the approximating distribution q needs to be chosen. Examples of the approximating distribution q are the Gaussian distribution when the noise is Gaussian (results in an objective similar to the ELBO objective popularized by Variational Autoencoders), learning the distribution, or non-parametric approximations when the noise is uniform. All these methods may yield a mechanism that computes $q(\hat{Y})$.

[0117] In an embodiment, distortion measure D is the L_2 norm, then the parameters of the encoder (Θ_e) and the decoder (Θ_d) may be learned by minimizing the following expression:

$$\mathcal{L}_R = \mathbb{E}_{X,Z}\left[R\|X - \hat{X}\|_2^2 - \log(q(\hat{Y}))\right] \tag{4}$$

[0118] FIG. **4** illustrates another set of operations **400** of controlled information flow for KD, which performs a training of the RDM **340** to mimic a TA. Specifically, FIG. **4** illustrates how the RDM **340** is trained. FIG. **4** illustrates the stage 1 of the training phase. In stage 2 of the training phase (illustrated in FIG. **5**), the RDM **340** is frozen and the student model **320** is trained. During this time, the knowledge is transferred from the teacher model **310** via the teacher embedding **410** and from the RDM network **340** via the RDM embedding **420**. Throughout the present disclosure, the teacher embedding **410** may correspond to the input embeddings **342** (of FIG. **3** and the RDM embedding **420** may correspond to the reconstructed embeddings **349** (of FIG. **3**).

[0119] In FIG. **4**, because the RDM **340** processes the teacher embeddings **410** (received from the teacher model **310**), the RDM **340** does not have to learn low-level feature extractors from the input, thus making the RDM **340** computationally cheaper compared to the TA.

[0120] As shown in FIG. **4**, the training of the RDM **340** includes calculating a 'mean squared error' (MSE) loss function and a rate loss function. A sum of these loss functions correspond to a total loss function. As known in the art, a loss function quantifies an error margin between a model's prediction and the actual target value. The loss function, also referred to as the error function, quantifies a difference between the predicted outputs of a machine learning model and actual target values. The loss function is a mathematical formulation of an objective of machine learning tasks.

[0121] The RDM embedding **420** may be used to train the student model **320**. In FIG. **5**, the student embeddings **520** are trained to mimic both the teacher embeddings **410** and the RDM embeddings **420** using the MSE loss function (i.e., "MSE Loss 1" and "MSE Loss 2" in FIG. **5**). In some embodiments, the MSE loss function may be replaced with other types of loss functions such as mean-absolute-error (MAE) loss function or smooth MAE loss function. FIG. **3** is a high-level representation of the system proposed in the disclosure, while FIG. **5** illustrates details as to how the proposed system is trained.

[0122] FIG. **5** illustrates another set of operations **500** of controlled information flow for KD, which performs a training of the student model **320** using the trained RDMs **340** and the teacher model **310**.

[0123] How the RDMs **340** mimic the TAs is described herein.

[0124] The TAs may limit the amount of information extracted from the input by using limited model capacity. The lesser the model capacity, the poorer the TA performance on a downstream task. On the other hand, the operations **500** of controlled information flow for KD may limit the amount of information (extracted from the teacher model's embedding) by passing the information through a rate constrained communication channel. If a higher constraint on the information is placed through the communication channel, reconstructed embeddings may have more distortion compared to the teacher model's and may perform poorly on the downstream task, just like the embeddings from a TA with a small model capacity. Thus, by choosing different Rs in the above expression (4), different TAs may be mimicked.

[0125] In a case where a classification model is distilled, an additional linear layer may be used to convert the reconstructed embeddings to logits. Let V represent the true classification label, $\hat{V}_T$ represent the teacher's output predictive distribution on the class labels, $\hat{V}_{RDM}$ be the same but as predicted by the RDM output. Then, the loss function used to train the RDM for classification is $\mathcal{L}$'RC:

$$\mathbb{E}_{X,Z}[\mathcal{L}_{CE}(V, \hat{V}) + \lambda_{KL} KL(\hat{V}_T \| \hat{V}_{RDM})] + \mathcal{L}_R \tag{5}$$

[0126] Here, $\mathcal{L}_{CE}$ is the cross-entropy loss, KL( ) is the Kullback-Leibler divergence, and λ_{KL} is weighting factor for KL( ) loss. FIG. **5** illustrates an example of the IBM **510**. FIG. **6** illustrates how the RDM **340** is trained for classification.

[0127] The IBM **510** is described below.

[0128] FIG. **5** illustrates an example of the IBM **510**. As a number of the RDMs **340** increases, the student model **320** may face a tendency to 'overfit' to stronger teacher models and stronger TAs. Thus, it may be important to constrain information from the student model **320** that is exposed to feedback from the teacher model **310** and the RDMs **340**. The IBM **510**, which is a module (or computer codes) about 'information bottleneck' in the student model **320**, may regularize the training. In an embodiment, the IBM **510** and the operations **300** of controlled information flow for KD may be performed together when the information is constrained both on the teacher model **310** and the student model **320**.

[0129] As described above, the RDMs **340** may be configured to control the flow of information from the teacher model **310** and mimic the TAs. The teacher model **310** and multiple RDMs **340** may provide feedback that may cause the student model to overfit and lead to poor performance. The overfit problem may be overcome by constraining the information from the student model **320** to be exposed to the feedback (i.e., partial feedback). In the operations **500** of controlled information flow for KD, similar to the teacher model **310**, the student model **320** may use another rate-constrained channel to constraint the information.

[0130] Further, this rate-constrained model (i.e., the student model **320** using another rate-constrained channel) may be present only when there is feedback to the student model **320**, i.e., only during a training of the student model **320**. However, unlike the case of the RDMs **340**, the rate-constrained model may not reconstruct the input to the rate constrained channel. Instead, the rate-constrained model may reconstruct the teacher model **310** or RDM embeddings. In information theory, the rate-distortion problem (where information about another random variable is retained in the compressed representation) is similar to an information bottleneck problem. Thus, the above rate-constrained model may be called as the IBM **510**.

[0131] Information bottleneck principle is a generalization to the rate-distortion problem. Given an input X_S (shown in FIG. **5**) and some random variable of interest U, the goal in the information bottleneck is to find a representation $\hat{U}$ (shown in FIG. **5**) that removes as much information about input X_S while retaining as much information about U. This is formulated as:

$$\min -I(U;\hat{U}) + \lambda_I I(X_S;\hat{U}) \tag{6}$$

[0132] Here, λ_1, is the Lagrange multiplier.

[0133] FIG. **5** shows the IBM in the student model. X_S represents the input, W represents the encoded representation, $\hat{W}$ is the noisy encoded representation, Z_S is the noise added during training only, and $\hat{U}$ represents the output of the IBM decoder.

[0134] When working with neural network-based encoders and decoders, both the mutual information terms in the above expression (6) may be intractable (not easily controlled or directed). The operations **500** of controlled information flow for KD may include variational approximations, resulting in an upper bound on the information bottleneck objective (6), which is represented as below:

$$\mathcal{L}_1 = \mathbb{E}_{X,Z}\left[D(U,\hat{U}) - \lambda_I \log(r(\hat{W}))\right] \tag{7}$$

where D represents a suitable distortion metric, $r(\hat{W})$ is the approximation of the true distribution $p(\hat{W})$.

[0135] In the IBM **510**, U is set to be the teacher model's (or RDM's) embedding. So, at the output, the IBM **510** attempts to reconstruct the teacher model's (or RDM's) embedding. In a case of a dimensionality mismatch, a projection layer is used. In an embodiment, a dedicated encoder for the IBM **510** may not be used and the student backbone model itself may become the encoder, i.e., $W=X_S$. In an embodiment, the decoder of the IBM **510** is a simple network of at most one or two linear layers. In an embodiment, the IBM may not be trained separately, instead the IBM may be trained along with the student model.

[0136] The IBM **510** may be used for a masked image modeling (MIM) as described below.

[0137] As known in the art, the MIM is a computer vision technique that involves predicting missing pixels in an image by using surrounding pixels as context. The MIM is often used in image inpainting, where missing or damaged parts of an image are filled in using information from the surrounding areas.

[0138] The MIM has been successful in pretraining large image models. In the MIM, an image is first converted into tokens using a pretrained tokenizer. Next, a masked version of the image is fed into the MIM encoder that attempts to predict the tokens of the masked parts. Masked tokens may be a random variable of interest for the IBM (U). The predicted tokens from the large image model encoder may be $\hat{U}$.

[0139] An upper-bound on the above expression (6) may be obtained by using the data processing inequality $I(X;\hat{U}) \leq I(X;X_M)$, wherein $\hat{U}$ is a function of X_M (say f):

$$\min -\mathbb{E}_{X_M}[\log p(U \mid f(X_M))] + \lambda_{MIM} I(X;X_M) - H(U) \tag{8}$$

[0140] Here, the second term $I(X;X_M)$ and the third term H(U) are constants regarding the parameters of the large image model encoder. The first term of the above expression (8) is the same as the training objective of the large image model. This shows that the training objective is an application of the information bottleneck principle for pretraining a large image model, except that unlike usual information bottleneck, the bottleneck is applied at the input, i.e., the masking. The amount of masking implicitly determines the information rate-constraint.

[0141] Unlike the MIM where the target U comes from the tokenizer, in the operations of controlled information flow for KD, the target U comes from the teacher model in the form of the embeddings of the teacher model. Further, while the masking applies an implicit rate-constraint, in the operations of controlled information flow for KD, an explicit rate-constraint is applied by minimizing an upper bound on the mutual information during the training.

[0142] A final loss function of the operations of controlled information flow for KD is described below.

[0143] In an embodiment, using the operations described above, a plurality of the RDMs **340** (N RDMs), which respectively correspond to different rate-constraints, may be trained. X is the input datapoint. The teacher model's embedding of X is U_T; U_n, Z_n (the embedding and noise in the nth RDM respectively). U_S corresponds to the student model's embedding. In an embodiment, the loss to match embeddings is the L_2 loss. Then, the loss function for training the student model **320** may be described as $\mathcal{L}_{CIFD}$:

$$\mathbb{E}_{X,Z_1^N,Z_S}\left[\|U_S-U_T\|_2^2+\sum_{n=1}^{n=N}\lambda_n\|U_S-U_n\|_2^2-\lambda_I\log r(\hat{W})\right] \tag{9}$$

[0144] Here, λ_n are weighting coefficients.

[0145] In a case of classification, the output predictive distribution of the teacher model **310** may be denoted as $\hat{V}_T$, $\hat{V}_S$ for the student model, $\hat{V}_{Tn}$ for the nth RDM, and the true label as V. The loss function for training the student model **320** may be described as:

$$\mathcal{L}'_{CIFD}=\mathbb{E}_{X,Z_1^N,Z_S}\Big[\lambda_{CE}\mathcal{L}_{CE}(V,\hat{V}_S)+\lambda_{KL}KL(\hat{V}_T\|\hat{V}_S)+\lambda_{KL}\sum_{n=1}^{n=N}\lambda_n KL(\hat{V}_n\|\hat{V}_S)\Big]+\mathcal{L}_{CIFD} \tag{10}$$

[0146] The operations of controlled information flow for KD of the present disclosure may be used for contrastive language-image pretraining (CLIP) as described below.

[0147] CLIP is a technique for training a pair of neural network models, one for image understanding and the other for text understanding, using a contrastive objective. CLIP has enabled multiple applications across multiple domains such as text-to-image generation and image capturing.

[0148] CLIP is a class of foundational models that are capable of embedding inputs from distinct modalities into a shared embedding space. In CLIP, a modality specific encoder processes the input from a specific modality and embeds the processed input into a shared embedding space. CLIP-like models have shown tremendous performance in zero-shot classification, object-detection, and retrieval. Further, the trained encoders have also proved instrumental in powering large multimodal models (LMMs) and generative models. Thus, distillation of these CLIP-like models has far-reaching applications especially in on-device generative artificial intelligence (AI).

[0149] In one embodiment, two modalities, Image ($\mathbb{I}$) and Language ($\mathbb{L}$), and a batch of B image-text pairs $\{(I^{(1)},L^{(1)}), \ldots, (I^{(B)},L^{(B)})\}$ are considered. $U_{\mathbb{I}}^{(b)}$ denotes the L_2-normalized embedding of the v-th image obtained from the image encoder. $U_{\mathbb{I}}^{(b)}$ denotes the same for the b-th text obtained from the language encoder. Then, the contrastive loss from an image to language embeddings is expressed as:

$$\mathcal{L}_{CL,\mathbb{I}\to\mathbb{L}}=\frac{-1}{B}\sum_{b=1}^{B}\log\frac{\exp\left(\langle U_{\mathbb{I}}^{(b)},U_{\mathbb{L}}^{(b)}\rangle/\tau\right)}{\sum_{k\in[B]}\exp\left(\langle U_{\mathbb{I}}^{(b)},U_{\mathbb{L}}^{(k)}\rangle/\tau\right)} \tag{11}$$

where $\langle\cdot,\cdot\rangle$ represents the inner-product between the two vectors. The loss to train CLIP-like models is represented as:

$$\mathcal{L}_{CL}=\mathcal{L}_{CL,\mathbb{I}\to\mathbb{L}}+\mathcal{L}_{CL,\mathbb{L}\to\mathbb{I}} \tag{12}$$

[0150] In contrastive loss, embeddings of a paired set of an image and a text are close to each other (e.g., high inner-product) compared to embeddings of a non-paired set of an image and a text.

[0151] Because the CLIP teacher model has modality-specific encoders, each encoder may have its own set of RDMs. Each of the set of RDMs **340** may be trained using the above expression (4). The embeddings for b-th image and text from the teacher are denoted as $U_{T,\mathbb{I}}^{(b)}$ and $U_{T,\mathbb{I}}^{(b)}$, respectively. The n-th RDM embeddings are defined as $U_{n,\mathbb{I}}^{(b)}$ and $U_{n,\mathbb{L}}^{(b)}$.

[0152] Similarly, because the student model **320** has modality-specific encoders, the student model **320** has modality-specific IBMs. $\hat{W}_{\mathbb{I}}$, $\hat{W}_{\mathbb{L}}$ (similar to $\hat{W}$ in FIG. **5**) for the image encoder IBM and the language encoder IBM, respectively. The output of the IBM decoder for the b-th image and text is denoted as $U_{S,\mathbb{I}}^{(b)}$ and $U_{n,\mathbb{I}}^{(b)}$, respectively. Then, the modality-specific CIFD loss for CLIP distillation is described as:

$$\mathcal{L}_{CIFD,\mathbb{I}}=\frac{1}{B}\sum_{b=1}^{B}\left[\left\|U_{S,\mathbb{I}}^{(b)}-U_{T,\mathbb{I}}^{(b)}\right\|_2^2+\sum_{n=1}^{n=N}\lambda_n\left\|U_{S,\mathbb{I}}^{(b)}-U_{n,\mathbb{L}}^{(b)}\right\|_2^2-\lambda_{I,\mathbb{I}}\log r_{\mathbb{I}}(\hat{W}_{\mathbb{I}})\right]$$

(13)

[0153] With all components in place, the final loss (to perform distillation of CLIP using the operations of controlled information flow for KD) is described as:

$$\mathcal{L}''_{CIFD}=\lambda_{CL}\mathcal{L}_{CL}+\mathcal{L}_{CIFD,\mathbb{I}}+\mathcal{L}_{CIFD,\mathbb{L}} \tag{14}$$

[0154] Here, λ_{CL} is weighting factor.

[0155] Operations for training the RDM **340** for classification tasks are described below in view of the example embodiment shown in FIG. **6**.

[0156] FIG. **6** illustrates another set of operations **600**, which includes training the RDMS **340** and the student model **320** for classification tasks.

[0157] In FIG. **6**, the input (e.g., an image) is passed through the teacher (backbone) model **310**. The teacher embedding **410** is passed through the encoder **344** of the RDM **340**, subject to noise (Z) and is reconstructed by the decoder **348** of the RDM **340**. In the case of classification, the RDM **340** is trained like a multi-task learning module. That is, the RDM **340** is tasked with reconstructing both the input embedding and projecting the reconstructed embedding to perform classification. The RDM **340** is trained in a similar fashion to the student model **320** during KD, i.e., there is feature distillation (reconstructing the teacher embedding), logit distillation to preserve the dark knowledge, and a supervised cross entropy loss component. The resulting loss function used to train the RDM is described as shown below:

$$\mathcal{L}'_R = \mathbb{E}_{X,Z}\Big[\lambda_{CE}\mathcal{L}_{CE}\big(V,\hat{V}\big) + \lambda_{KL}KL\big(\hat{V}_T \| \hat{V}_{RDM}\big) + R\big\|X-\hat{X}\big\|_2^2 - \log\big(q(\hat{Y})\big)\Big] \tag{15}$$

[0158] That is, as shown in FIG. **6**, the total loss is equal to a sum of the MSE loss, the rate loss, KL Div. loss (that is, Kullback-Leibler divergence loss), and the CE loss (that is, cross entropy loss).

[0159] FIG. **7** illustrates another set of operations **700**, which include training the student model **320** for classification. FIG. **7** illustrates how the student models **320** are trained for classification in the presence of the RDM **340** and the IBM **510**. FIG. **7** also illustrates the teacher FC **610**, the RDM FC **620**, and a student FC **710**.

[0160] In an embodiment, there is one RDM **340**. As illustrated in FIG. **7**, the input image is passed through the teacher (backbone) model **310** and the obtained embedding is passed through the RDM **340** to get the reconstructed embedding from the RDM **340**. Then, the RDM **340** provides a predictive distribution (like a TA) and the teacher model's predictive distribution is obtained.

[0161] The input image is also passed through the student model **320** and the IBM **510** to obtain the student embedding **520** that is then subject to feature distillation, i.e., the loss between the student embedding **520** and the teacher embedding **410** and the loss between the student embedding **520** and the RDM embedding **420** are computed. In a case where a dimension of the student model's embedding does not match with a dimension of the teacher model's embedding or the RDM's embedding, a small trainable projector network is used. Finally, the output predictive distribution of the student model **320** is subject to both the classification loss and the KL divergence losses with respect to the teacher model's distribution and the RDM's distribution.

[0162] So, the set of operations **700** combines feature distillation and the output logit distillation along with the RDM **340**. FIG. **7** shows one RDM as an example. However, the disclosure is not limited to one RDM **340**. Embodiments of the disclosure may include multiple RDMs **340** (for example, as illustrated in FIG. **4**).

[0163] The loss function used in the set of operations **700** is:

$$\mathcal{L}'_{CIFD} = \mathbb{E}_{X,Z_1^N,Z_S}\Big[\lambda_{CE}\mathcal{L}_{CE}\big(V,\hat{V}_S\big) + \lambda_{KL}KL\big(\hat{V}_T \| \hat{V}_S\big) + \lambda_{KL}\sum_{n=1}^{n=N}\lambda_n KL\big(\hat{V}_n \| \hat{V}_S\big) + \|U_S-U_T\|_2^2 + \sum_{n=1}^{n=N}\lambda_n\|U_S-U_n\|_2^2 - \lambda_I \log r(\hat{W})\Big] \tag{16}$$

[0164] That is, as illustrated in FIG. **7**, the total loss is equal to a sum of the MSE loss 1, the MSE loss 2, the rate loss, the KL div. loss 1, the KL div. loss 2, and the CE loss.

[0165] FIG. **8** illustrates examples of practical applications of the operations described above. As illustrated in FIG. **8**, a system **800** (e.g., a cloud system, a computer system) includes the first electronic device **101** (e.g., a user terminal) and the server **108**. FIG. **1** illustrates example components of the first electronic device **101**. FIG. **2** illustrates example components of the server **108**.

[0166] As illustrated in FIG. **8**, the server **108** may perform at least one of the operations **300**, **400**, **500**, **600**, and **700** that are described above. After the server **108** preforms the at least one of these operations, the output of the operations, which is a trained student model **320** (i.e., a student model **320** trained with the teacher model **310** and the RDM **340** at the server **108**), is transferred to the first electronic device **101** (operation **802**). In one embodiment, the first electronic device **101** transmits a request (operation **804**) for the trained student model **320** to the server **108**. Based on the request from the first electronic device **101**, the server **108** may forward the trained student model **320** to the first electronic device **101**. In one embodiment, the server **108** may forward the trained student model **320** to the first electronic device **101**, although there is no prior request from the first electronic device **101**.

[0167] FIG. **9** illustrates example set of operations **900** in accordance with some embodiments of the disclosure. At operation **902**, the server **108** obtains training data. At operation **904**, the server **108** obtains a teacher model **310**. Examples of the teacher model **310** are a pretrained teacher model **310** and a new teacher model **310** to be trained by the server **108**.

[0168] At operation **906**, the server **108** trains at least one RDM **340**, for example, as illustrated in FIG. **4** and FIG. **6** and as described above. Also, FIG. **10** illustrates sub-operations of operation **906**.

[0169] At operation **908**, the server **108** trains a student model **320** using the trained at least one RDM **340**, optionally, with the IBM **510**, for example, as illustrated in FIG. **5** and FIG. **7** and as described above. Also, FIG. **11** illustrates sub-operations of operation **908**.

[0170] As shown in FIG. **9**, the server **108** performs operations **902** to **908** during a training phase.

[0171] At operation **909**, the server **108** transmits the trained student model **320** to the first electronic device **101** (e.g., the user terminal). Operation **909** may be equal to or correspond to operation **802** of FIG. **8**. In some embodiment, the server **108** receives a command from a user through the user interface **210** shown in the display **208**. The user may select the trained student model through the user interface **210** and instruct to transmit the selected trained student model from the server **108** to the first electronic device **101**. In some embodiments, the server **108** may transmit the trained student model **320** to the first electronic device **101** even without any request from the first electronic device **101**.

[0172] At operation **910**, the first electronic device **101** obtains data for prediction. For example, the data for prediction are data for chatbot services (representing users' queries), data for chat assistants (representing users' queries), data for analysis, and data for business intelligence.

[0173] At operation **912**, the first electronic device **101** generates the prediction using the trained student model **320**. That is, the first electronic device **101** provides the obtained data to the trained student model **320** and obtains results of the prediction (e.g., the chatbot services (outputting answers to the users' queries), chat assistants (outputting answers to the users' queries), the analysis, or the business intelligence) from the trained student model **320**.

[0174] At operation **914**, the first electronic device **101** outputs the generated prediction through a user interface of the first electronic device **101**. For example, a user of the first electronic device **101** receives, via the user interface of the first electronic device **101**, chat responses generated by the trained student model **320**.

[0175] As shown in FIG. **9**, the first electronic device **101** performs operations **910** to **914** during an inference phase.

[0176] FIG. **10** illustrates example sub-operations of operation **906** (training the at least one RDM **340**), which are illustrated in FIG. **4** or FIG. **6**.

[0177] At operation **1000**, the server **108** calculates a first loss function based on the teacher embeddings **410** and the RDM embeddings **420**.

[0178] At operation **1002**, the server **108** calculates a second loss function based on an output of the encoder **344** and noise.

[0179] At operation **1003**, optionally, the server **108** calculates other loss functions.

[0180] At operation **1004**, the server **108** determines a total loss function by adding (at least) the first loss function and the second loss function.

[0181] At operation **1006**, the server **108** trains the at least one RDM **340** using the total loss function determined at operation **1004**.

[0182] FIG. **11** illustrates example sub-operations of operation **908** (training the student model **320**), which are illustrated in FIG. **5** or FIG. **7**.

[0183] At operation **1100**, the server **108** calculates a first loss function based on the teacher embeddings **410** and the student embeddings **520**.

[0184] At operation **1102**, the server **108** calculates a second loss function based on the RDM embeddings **420** and the student embeddings **520**.

[0185] At operation **1104**, the server **108** calculates a third loss function based on an output of an encoder of the IBM **510** and noise.

[0186] At operation **1106**, optionally, the server **108** calculates other loss functions.

[0187] At operation **1108**, the server **108** determines a total loss function by adding (at least) the first loss function, the second loss function, and the third loss function.

[0188] At operation **1110**, the server **108** trains the student model **320** using the total loss function determined at operation **1108**.

[0189] One or more embodiments as set forth herein may be implemented as software including one or more instructions that are stored in a storage medium that is readable by a machine. For example, a processor of the machine may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a complier or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term "non-transitory" simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

[0190] According to an embodiment, a method according to one or more embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store, or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

[0191] According to one or more embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities. According to one or more embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. In some embodiments, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to one or more embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to one or more embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

[0192] According to one or more embodiments, in non-volatile storage medium storing instructions, the instructions may be configured to, when executed by at least one processor, cause the at least one processor to perform at least one operation. The at least one operation may include displaying an application screen of a running application on a display, identifying a data input field included in the application screen, identifying a data type corresponding to the data input field, displaying at least one external electronic device, around the electronic device, capable of providing data corresponding to the identified data type, receiving data corresponding to the identified data type from an external electronic device selected from among the at least one external electronic device through a communication circuit, and entering the received data into the data input field.

[0193] The embodiments of the disclosure described in the disclosure and the drawings are only presented as specific examples to easily explain the technical content according to the embodiments of the disclosure and help understanding of the embodiments of the disclosure, not intended to limit the scope of the embodiments of the disclosure. Therefore, the scope of one or more embodiments of the disclosure should be construed as encompassing all changes or modifications derived from the technical spirit of one or more embodiments of the disclosure in addition to the embodiments disclosed herein.

What is claimed is:

1. A computer-implemented method performed by an electronic device configured to distill knowledge from a teacher model and transfer the distilled knowledge to a student model, the computer-implemented method comprising:

obtaining training data;

obtaining the teacher model;

training at least one rate distortion module (RDM) that is not a teacher assistant (TA) model;

training the student model using the trained at least one RDM; and

transmitting the trained student model to a first electronic device.

2. The computer-implemented method of claim **1**, wherein the transmitting the trained student model to a first electronic device, comprises:

receiving, via a user interface of a display in the electronic device, a user's first command to select the trained student model,

receiving, via the user interface, the user's second command to transmit the trained student model to the first electronic device, and

transmitting, based on the user's second command, the trained student model to a first electronic device.

3. The computer-implemented method of claim **1**, wherein the teacher model is a large language model (LLM) and the student model is a small language model (SLM).

4. The computer-implemented method of claim **1**, wherein the teacher model is a pretrained model that is pretrained by an external device.

5. The computer-implemented method of claim **1**, wherein the teacher model is a large language model trained by the electronic device.

6. The computer-implemented method of claim **1**, wherein the first electronic device is a user terminal.

7. The computer-implemented method of claim **1**, the training the at least one RDM, comprises:

calculating a first loss function based on teacher embeddings and RDM embeddings;

calculating a second loss function based on an output of an encoder and noise;

determining a total loss function by adding at least the first loss function and the second loss function; and

training the at least one RDM using the determined total loss function.

8. The computer-implemented method of claim **7**, further comprising calculating other loss functions,

wherein the determining the total loss function by adding at least the first loss function and the second loss function comprises determining the total loss function by the first loss function, the second loss function, and the other loss functions.

9. The computer-implemented method of claim **1**, the training the student model using the trained at least one RDM, comprises:

calculating a first loss function based on teacher embeddings and student embeddings;

calculating a second loss function based on RDM embeddings and the student embeddings;

calculating a third loss function based on an output of an encoder of an information bottleneck module (IBM) and noise;

determining a total loss function by adding at least the first loss function, the second loss function, and the third loss function; and

training the at least one RDM using the determined total loss function.

10. The computer-implemented method of claim **8**, further calculating other loss functions,

wherein the determining the total loss function by adding at least the first loss function, the second loss function, and the third loss function comprises determining the total loss function by the first loss function, the second loss function, the third loss function, and the other loss functions.

11. An electronic device configured to distill knowledge from a teacher model and transfer the distilled knowledge to a student model, the electronic device comprising:

at least one memory;

a display displaying a user interface; and

at least one processor operatively connected with the at least one memory and the display;

wherein the at least one processor is configured to perform:

obtaining training data;

obtaining the teacher model;

training at least one rate distortion module (RDM) that is not a teacher assistant (TA) model;

training the student model using the trained at least one RDM; and

transmitting the trained student model to a first electronic device.

12. The electronic device of claim **11**, wherein the at least one processor is further configured to perform:

receiving, via the user interface, a user's first command to select the trained student model,

receiving, via the user interface, the user's second command to transmit the trained student model to the first electronic device, and

transmitting, based on the user's second command, the trained student model to a first electronic device.

13. The electronic device of claim **11**, wherein the teacher model is a large language model (LLM) and the student model is a small language model (SLM).

14. The electronic device of claim **11**, wherein the teacher model is a pretrained model that is pretrained by an external device.

15. The electronic device of claim **11**, wherein the teacher model is a large language model trained by the electronic device.

16. The electronic device of claim **11**, wherein the first electronic device is a user terminal.

17. The electronic device of claim **11**, wherein the at least one processor is further configured to perform:

calculating a first loss function based on teacher embeddings and RDM embeddings;

calculating a second loss function based on an output of an encoder and noise;

determining a total loss function by adding at least the first loss function and the second loss function; and

training the at least one RDM using the determined total loss function.

18. The electronic device of claim **17**, the at least one processor is further configured to perform:

calculating other loss functions, and

determining the total loss function by the first loss function, the second loss function, and the other loss functions.

19. The electronic device of claim **11**, wherein the at least one processor is further configured to perform:

calculating a first loss function based on teacher embeddings and student embeddings;

calculating a second loss function based on RDM embeddings and the student embeddings;

calculating a third loss function based on an output of an encoder of an information bottleneck module (IBM) and noise;

determining a total loss function by adding at least the first loss function, the second loss function, and the third loss function; and

training the at least one RDM using the determined total loss function.

20. The electronic device of claim **18**, wherein the at least one processor is further configured to perform:

calculating other loss functions, and

determining the total loss function by the first loss function, the second loss function, the third loss function, and the other loss functions.

* * * * *